The School of Mathematics

Density-Aware Graph Geodesic Stress Testing on a Learned Macroeconomic Manifold

by

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Abstract

[abstract]

Acknowledgments

[acknowledgments]

Own Work Declaration

[own work declaration]

AI disclosure

[see guidance document]

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1 Introduction

1.1 motivation

Financial stress testing defines a methodology which aims to understand "the extent to which a portfolio, individual institution, or a group of institutions is capable of withstanding external shocks" [7]. For example, a bank may ask a fund manager "what will happen to your portfolio if we experience an economic shock similar to COVID-19 next year". The purpose of this is to ensure that individuals and institutions are able to stay afloat during times of economic turmoil; as opposed to defaulting and needing baled out. While the specific economic stress point is important, what is potentially equally important is the path the economy takes to get there. This is because in many portfolios and institutions, the final value may change along the way, due to risk limits or margin calls which would potentially mean a portfolio would have to be rebalanced.

1.2 motivation notes

Understand what happens in stressed economic scenarios. For example a portfolio may loose £1m if sp500 falls 5% and interest rates rise 1%? This number is often implicitly computed under the assumption that all other risk factors are set to zero. This is generally not justified because the non-stressed factors conditioned on the stress factors would be set to zero. Even if the non-stress factors are set to their conditional expected values, under the stress situation, then the reported P&L may be non-zero because of the convexity of the portfolio value functions, especially the case when there are options in the portfolio. Portfolio managers must work within risk constraints. A constraint being VaR for example. [16]

Use PCA to create stress situations by evolving along the "least energy" direction in (principle component space) in order to arrive at a certain shock value for one variable, while also moving the rest of the variables. [27] This could be interesting to look at using diffusion coordinates

Uses PCA to generate multidimensional stress scenarios [14]

Stress testing models are not robust because relationships between factors tend to break down during crises [1]

The 47 countries are forced to do an in-depth analysis of their finance sector. This includes stress tests. These are the largest economies in the world. [19]

In the IMF's hand book on stress testing "A Guide to IMF Stress Testing" (2014), they outline how they use PCA in their contingent claims analysis [11]

In the IMF's 2021 Financial Sector Assessment Program (FSAP) Review, background paper on quantitative methods, they use PCA for their growth at risk (GaR) metric. They say that the non-linear relationship between GDP growth and financial conditions is estimated using quantile regression. [2]

IMF paper on West African Economic and Monetry Union shows they use PCA for growth at risk and inflation at risk [18]

In contrast with the mainstream macroeconomic approach with representative agents and equilibrium relationships, stress tests try to capture tail events, which usually entail sudden shifts in markets, so historical distributions may not adequately capture the non-linear dynamics under stress ([6], 2012).

More agent based models for stress testing, how shocks actually travel through system, rather than what will the value of xyz be in stressed state. And Financial stress testing gained momentum among financial regulators after the GFC because risk measure based on historical relationships failed to provide adequate insight. [5]

In Growth at Risk: Concept and Application in IMF Country Surveillance [25], they talk a lot about how economic growth reacts nonlinearly to shocks. They say GaR centers on forcasting probability distribution of future GDP growth in a way that allows for nonlinearity and state dependence. It says that it captures the rich nonlinear interaction between shicks, financial conditions and economic outcomes predicted by theory. They say quantile regression finds nonlinear relationships between macrofinancial conditions and future GDP growth.

Step 1: Select macrofinancial variables and construct partitions (partitions are constructed with PCA) Step 2: Quantile regression on the chosen variables/partitions, this allows for nonlinearities, but we are working with linear partitions of potentially nonlinear substructures. Step 3: Derive the conditional future growth distributions. fit t-skew distributions to predicted values of the estimated conditional quantiles regressions. Actually seems quite cool. more steps... This is a framework for the countries doing the IMF stress testing to use [25]

"As noted in the introduction, applying diffusion maps directly to time series data may be problematic since the data is not sampled uniformly and one should ideally take into account the temporal relationships in the data" [3] "There, derivatives portfolios are considered and the risk factors are low-dimensional, as is usually the case in stress testing." [3]

1.3 after notes

The question then becomes, how do we generate economically plausible stress paths? Before we tackle this question, we introduce something called the manifold hypothesis. We assume that the all of the observed variables in the economy are driven by a smaller number of underlying, unseen factors. This then suggests that the observed economic variables may concentrate near a lower dimensional structure. This motivates the use of dimensionality reduction schemes to try to describe the economy in a smaller number of dimensions. Linear dimension reduction methods, like Principle Component Analysis (PCA), assume that the data can be described by a linear subspace. PCA identifies the linear direction of maximal variance and uses the leading maximal directions of variance to generate a lower-dimensional subspace which can capture a lot of the variance in the data. A problem with this is that the relationship between macroeconomic varaibles, and themselves and the underlying factors may not be linear. This motivates the use of diffusion maps for a dimension reduction scheme, which was introduced by [10] as a "geometrically faithful dimensionality reduction technique" [3].

Even when using diffusion maps, the challenge of generating stress paths is not trivial. In [3], they use predefined stress paths for a subset of the macroeconomic variables to find a conditional distribution on the rest of the variables, and hence can generate stress paths. This is done by running a Kalman filter on the diffusion coordinates. In this dissertation, we intend to find a way to interpolate between two endpoints without any information about the interior points.

Talk about the combinations of macroeconomic variables which are not feasible via linear methods.

- What is stress testing?
- Introduce the manifold hypothesis
- Method reduction methods
- Diffusion maps
- In [3] they use linear paths in diffusion space, we want to beat that
- What can this be used for?

Limitations of linear interpolation through Euclidean space; it gives unrealistic combinations of variables.

Incoporate JDKF into motivation - argument runs that Baker has diffusion maps but linearly interpolates; can we find density aware paths

Each month gives many observations, but these could be represented using far fewer dimensions.

- learn low-dimensional representation of macro-financial states;

- construct stress transitions that respect the learned geometry and historical density;
- determine whether temporal regimes emerge in this latent representation.

1.4 Research questions

In this dissertation, we first try to find out whether diffusion maps can be used to recover stable and interpratible low-dimensional representations of macroeconomic-financial states. Then using this coordinate system, can we develop a novel method for interpolating between two endpoints in diffusion space whose points contain macro-financial combinations which are highly historically supported? The interpretatoin of these trajectories would be a stress path which passes through regions of macro-financial space which are highly supported by historical observations.

Main questions:

- Can diffusion maps recover a stable and interpretable low-dimensional representation of macro-financial conditions?
- Do graph-based and density-aware stress paths remain in more historically plausible regions than direct linear interpolation?

Extension questions:

- Do hidden Markov regimes fitted in diffusion space correspond to recognised economic regimes, including NBER recessions?
- Do density-aware geodesics pass through the inferred regimes in a temporally and economically coherent way?

1.5 contributions

In this dissertation, we construct a diffusion map from monthly macroeconomic and industry performance observations going back to 1964. We interpret each diffusion coordinates economic interpretation using non-linear distance correlation metrics and eocnomic reasoning. We estimate historical support for diffusion coordiantes via a density proxy. Using the density proxy, we introduce a density-aware graph cost for constructing stress transitions. We compare our density-aware path with linaer interpolations I think this will need to be explain/justified more above. Linear interpolation makes sense in PCA space because everything is linear and the space is defined. Whereas diffusion “space” is a concept, we have got a distrete approximation of it and unpenalised graph paths across four historical scenarios. We additionally employ a define a lifting procedure which allows for diffusion coordinate paths to be interpreted as macroeconomic variables.

Have done:

- Implementation and numerical validation of a diffusion-map pipeline;
- Construction of density-aware graph geodesics for macroeconomic stress testing;
- Lifting of latent stress paths into economically interpretable variables

Aim to do:

- Fitting of a Gaussian HMM to diffusion coordinates rather than the original high-dimensional observations;
- Joint evaluation of geometric paths, historical density, and temporal regimes.

1.6 Findings

We find that the density-aware paths generally improve the minimum and weighted mean historical support, compared to linear interpolation in diffusion space and ordinary graph paths.

2 Background

2.1 High-dimensional state vectors and the manifold hypothesis

Here we introduce the notation used throughout this dissertation and expand on the manifold hypothesis discussed in Section 1. We define a macroeconomic-financial state variable as

$$z_t = [z_t^1, \ldots, z_t^D]^\top \in \mathbb{R}^D,$$

which represents a variety of different macroeconomic and industry variables at time t. For example, z_t^1 may represent the unemployment rate and z_t^D may represent the previous month returns of a financial asset at time t. Although D may be very large, the collection of observed states may concentrate near a lower-dimensional set

$$\mathcal{M} \subset \mathbb{R}^D,$$

where $\dim(\mathcal{M}) \ll D$. In this dissertation, we assume the existence of some sort of non-linear manifold $\mathcal{M}$, but it is never explicitly calculated or estimated. The intrinsic geometry of $\mathcal{M}$ is learned via the diffusion mapping. citation for this theory. where did it come from? Can I just explain it well enough as a theory? The Manifold Hypothesis is Assumption 2 of [3]. In their introduction, they just say that they assume that the covariates and responses $(x(t), y(t))$ are nonlinear functions of a true, possibly lower-dimensional, underlying state of the system $\theta_t \in \mathbb{R}^d$. We don't ever use these θ_t explicitly, so is it worth introducing them rather than just explaining the concept?

Do I need a link to the next part of the background to this part? Like I did in my intro? I think right now they seem slightly disconnected. Should I say "The fact that the manifold may be non-linear motivates us to use diffusion maps which we outline below"?

2.2 Latent diffusions and spectral coordinates

I don't think that this part is necessary. Maybe this is where you can explain how the eigenvalues approach the eigenfunctions of the Laplace-Beltrami operator on the underlying manifold.

2.3 Diffusion maps

This outlines how we actually build the diffusion maps, rather than the theory above. Include an example of what diffusion coordinates actually do. I think the current version is far too technical and only shows how to build diffusion map. Need to give background

2.3.1 Diffusion maps motivation notes

Introduced by Coifman and Lafon [10], diffusion maps is a geometrically faithful dimensionality reduction technique. Using the eigenfunctions of Markov matrices, explain eigenfunction we can generate efficient representations of complex geometric structures.

Graph methods play a big role in modelling of social networks. Proposed a new model of a network where probability of interactions are based on physical distance. [17]

We can combine graph methods with Markov chain techniques which has proven useful in finding structures of complex geometries (like for spectral clustering from NMD, but we use Laplacian instead of Markov matrix which have the same eigenvalues). Can use the distance between the probabilities of transitions between data points as a metric, then this can be used to induce class labes. This is useful for non-linear geometries [10]

In spectral clustering, a Markov chain is constructed over the graph of the data, and the sign of the values at the first non-constant eigenvector of the corresponding transition matrix is used for finding clusters and computing cuts [29]. The normalised Laplacian and the transition matrix have the same eigenvectors.

PageRank uses the information from the stationary distribution of a random walk on the link structure of the web in order to rank web documents in terms of importance [10], [29], [8] [23]. I haven't read [8] or [23], they are citations within [10]. Probably won't use either, but ask James.

There are other variations which also use the principle eigenvalues of the Markov matrix on the graph to rank pages. [20] uses a different graph construction so a different random walk on the links, but the sites are still ranked by the values in the principle eigenvalue of the Markov matrix.

This approach is generalised to higher-order eigenvectors [10]

The following are example of kernel eigenmap methods. Diffusion maps are the general case of this. Locally linear embedding: For each node i, find weights for edges between i and j such that j is one of k nearest neighbour of node i and that the weights sum to 1. Uses leading eigenvectors as orthogonal coordinates. [26]

Hessian eigenmaps improve on LLE. Didn't read this one either, but thought I'd keep the reference incase I need to read it. [12]

Laplacian eigenmaps for dimension reduction. They construct the graph in a way I think would be quite interesting, they choose which nodes are connected by either distance or kNN and then they weight them by the heat kernel, this could be useful for the problem of the Markov bridges all going to the same place. [4]

2.3.2 Diffusion maps current background

Diffusion maps are a nonlinear dimensionality-reduction technique which use local similarities between observations to recover the geometry of a data set [10]. Suppose that

$$\mathcal{Z} = \{z_1, \dots, z_N\} \subset \mathbb{R}^D$$

is a collection of observations sampled from, or concentrated near, a lower-dimensional manifold

$$\mathcal{M} \subset \mathbb{R}^D.$$

The method constructs a weighted graph on the observations and uses the spectral properties of a random walk on this graph to obtain intrinsic coordinates for the data.

The first step is to calculate the pairwise squared Euclidean distances

$$\delta_{ij} = \|z_i - z_j\|^2.$$

These distances are converted into affinities using the Gaussian kernel

$$W_{ij} = \exp\left(-\frac{\|z_i - z_j\|^2}{\varepsilon}\right), \tag{2.1}$$

where the bandwidth $\varepsilon > 0$ controls the scale over which two observations are regarded as neighbours. Mention the fact that this is different to Mahalanobis distance used in [3]. Mahalanobis is not sensitive to nonlinearity and correlation. You have to estimate the correlations to use this; with $\tilde{Z} \in \mathbb{R}^{N\times D}$ this would introduce errors. Using the quantile data, we don't have to worry about the nonlinearity because it makes everything Gaussian, but we need to explain the correlation thing. The matrix

$$W = (W_{ij})_{i,j=1}^N$$

is symmetric and contains values between zero and one. Nearby observations have affinity close to one, whereas the affinity between distant observations is close to zero. Since each observation has zero distance from itself, $W_{ii} = 1$.

The weighted degree of observation i is defined by

$$q_i = \sum_{j=1}^{N} W_{ij}. \tag{2.2}$$

Unlike the degree of an unweighted graph, q_i does not simply count the number of adjacent vertices. It measures the total affinity of z_i to the sample. Consequently, a large value of q_i indicates that z_i has many nearby observations, or several observations to which it has strong affinity. Conversely, a value close to one indicates that almost all of the degree is contributed by the diagonal entry $W_{ii} = 1$, and hence that z_i is nearly isolated at the chosen bandwidth. The degree therefore acts as a kernel-based measure of local sampling density.

If the observations are sampled non-uniformly from $\mathcal{M}$, densely sampled regions may dominate the resulting random walk. Diffusion maps allow this effect to be controlled through the density-normalised affinity matrix

$$W_{ij}^{(\alpha)} = \frac{W_{ij}}{q_i^{\alpha} q_j^{\alpha}}, \qquad \alpha \in [0, 1]. \tag{2.3}$$

The parameter α determines how strongly the empirical sampling density is removed. The choice $\alpha = 0$ leaves the original affinities unchanged, whereas $\alpha = 1$ removes the leading influence of non-uniform sampling density and, under standard asymptotic conditions, yields an approximation to the Laplace–Beltrami operator associated with the underlying manifold [10]. Intermediate choices retain some dependence on the sampling distribution.

A second degree vector is then formed from the normalised affinities:

$$d_i^{(\alpha)} = \sum_{j=1}^{N} W_{ij}^{(\alpha)}. \tag{2.4}$$

Writing

$$D_{\alpha} = \operatorname{diag}\left(d_1^{(\alpha)}, \ldots, d_N^{(\alpha)}\right),$$

the diffusion transition matrix is

$$P = D_{\alpha}^{-1} W^{(\alpha)}, \qquad P_{ij} = \frac{W_{ij}^{(\alpha)}}{\sum_{k=1}^{N} W_{ik}^{(\alpha)}}. \tag{2.5}$$

Each row of P sums to one, so P can be interpreted as the transition matrix of a Markov chain on the data. In particular, P_{ij} is the probability of moving from observation z_i to observation z_j in one step. The entries of P^t give the corresponding transition probabilities after t steps.

Although P is generally not symmetric, it is similar to the symmetric matrix

$$S = D_{\alpha}^{-1/2} W^{(\alpha)} D_{\alpha}^{-1/2}. \tag{2.6}$$

I think I should leave this section out of the background. This is just an implementation choice which goes in methodology Indeed,

$$P = D_{\alpha}^{-1/2} S D_{\alpha}^{1/2}.$$

The matrices P and S therefore have the same eigenvalues. Since S is symmetric, its eigenvectors are orthogonal and its eigendecomposition is numerically more stable than directly diagonalising P. If

$$S u_k = \lambda_k u_k,$$

then the corresponding right eigenvector of P is

$$\phi_k = D_{\alpha}^{-1/2} u_k.$$

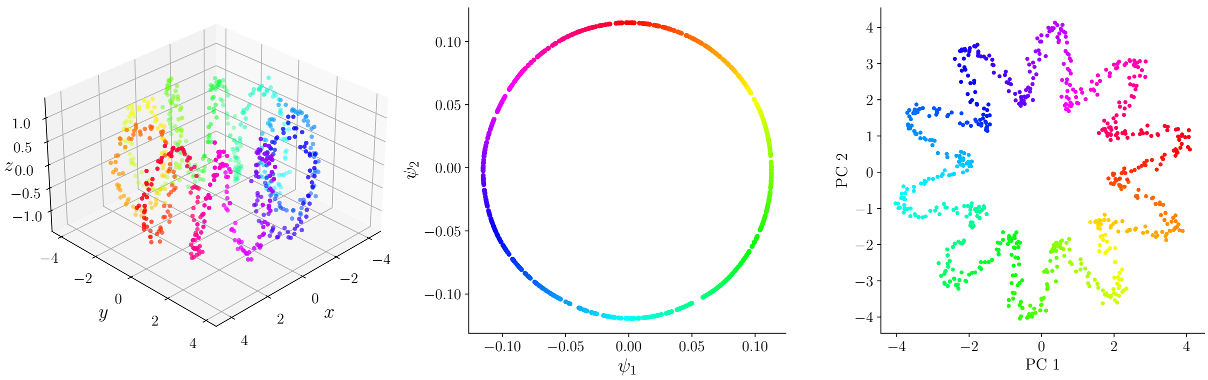


Figure 1: Caption

The eigenvalues are ordered as

$$1 = \lambda_0 \geq \lambda_1 \geq \lambda_2 \geq \cdots ,$$

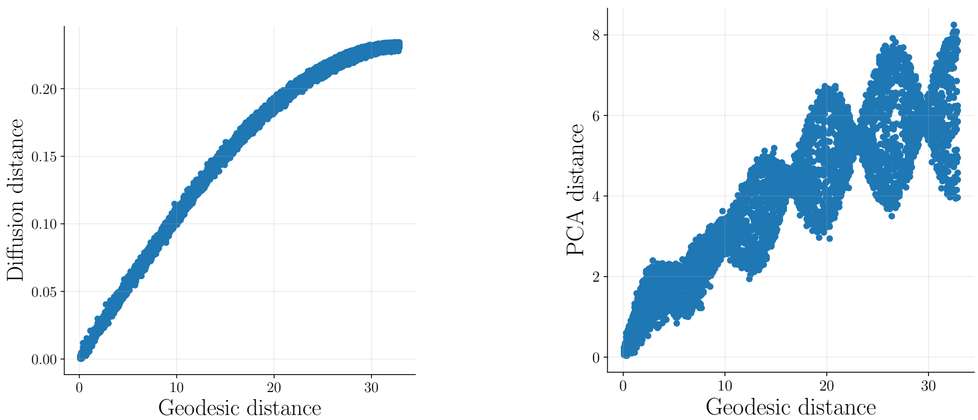
where the first eigenvector ϕ_0 is constant and carries no geometric information. The diffusion-map embedding at diffusion time t is therefore defined using the leading non-trivial eigenpairs:

$$\Psi_t(z_i) = \big(\lambda_1^t \phi_1(i), \ldots, \lambda_r^t \phi_r(i)\big) \in \mathbb{R}^r. \tag{2.7}$$

The eigenvalue factors suppress rapidly decaying modes, while the eigenvectors describe the dominant directions of diffusion through the data. With the complete eigendecomposition, Euclidean distance in diffusion-coordinate space is equal to diffusion distance:

$$D_t^2(z_i, z_j) = \sum_{k \geq 1} \lambda_k^{2t} \left(\phi_k(i) - \phi_k(j)\right)^2 .$$

Truncating this expansion after r terms provides a low-dimensional approximation that retains the most persistent diffusion modes. How does this relate to the geodesic distances?



(a) Diffusion coordinates

(b) PCA

Figure 2: Scatter plots of the distance between pairs in the lower-dimensional subspace and the geodesic distance between the same pairs, for both diffusion map and PCA of the Toroidal helix data points.

Under suitable assumptions on the sampling process, together with an appropriate relationship between the sample size and the bandwidth, the graph-based operator converges to a differential operator on $\mathcal{M}$. Correspondingly, its eigenvectors converge to evaluations of the eigenfunctions of the limiting operator at the sampled observations. This provides the theoretical basis for interpreting diffusion coordinates as empirical approximations to intrinsic spectral coordinates of the underlying system. In the latent-dynamical setting considered in Section 2.2, this connection motivates the use of diffusion maps to obtain data-driven coordinates for an otherwise unobserved state space.

The bandwidth ε is central to the construction. If ε is too small, the off-diagonal affinities become negligible, so that

$$W \approx I$$

and many observations are effectively isolated. In this regime, the diffusion operator describes a poorly connected collection of local components. If ε is too large, most affinities approach one, so that

$$W \approx \mathbf{1}\mathbf{1}^{\top},$$

and the kernel loses its ability to distinguish local geometric structure. An appropriate bandwidth must therefore be sufficiently large to connect the observations while remaining sufficiently local to preserve the geometry of the data.

The following comes from [28] (Figure 3, Equation 26). Might be worth mentioning "A popular choice for practitioners is to choose ε as the median of all distances between data points" [3]. One diagnostic for the bandwidth is the kernel mass

$$T(\varepsilon) = \sum_{i,j=1}^{N} \exp\left(-\frac{\|z_i - z_j\|^2}{\varepsilon}\right). \tag{2.8}$$

For observations sampled from a sufficiently regular d-dimensional manifold, the kernel mass is expected to satisfy

$$T(\varepsilon) \approx C\varepsilon^{d/2}$$

over an appropriate range of local scales. It follows that

$$\frac{d \log T(\varepsilon)}{d \log \varepsilon} \approx \frac{d}{2}. \tag{2.9}$$

A stable region in the log–log slope can therefore provide information about both the intrinsic dimension and a suitable range of bandwidths. In finite, noisy, and non-uniformly sampled data, however, such a scaling region may be weak or absent. Kernel-mass scaling is therefore best treated as a diagnostic and considered alongside graph connectivity, the degree distribution, spectral stability and the quality of the resulting coordinates.

The following comes from [9], Subsection 3.2 The embedding dimension r determines how many non-trivial diffusion modes are retained. A common diagnostic is the presence of a spectral gap. If

$$\lambda_r - \lambda_{r+1}$$

is large relative to neighbouring gaps, the first r non-trivial eigenvectors represent comparatively persistent modes, whereas subsequent eigenvectors decay more rapidly and may primarily reflect fine-scale variation or noise. Spectral gaps do not provide an infallible choice of dimension, particularly when eigenvalues occur in near-degenerate groups. The final dimension should therefore also be assessed through stability under nearby bandwidths, preservation of relevant geometric structure, and the requirements of the downstream analysis.

Include the example here? I am not sure how the example will exactly fit in.

2.4 Graph geometry and shortest paths

The diffusion coordinates are a way to represent our data in a lower-dimensional structure. And as shown above, the short scale distances in diffusion space highly correlate to distances in the intrinsic geometry. I.e. for short distances, diffusion distance and geodesic distances are tied together. This leads us to wonder whether paths through diffusion coordinates via short jumps correspond to geodesic paths between points in the underlying geometry.

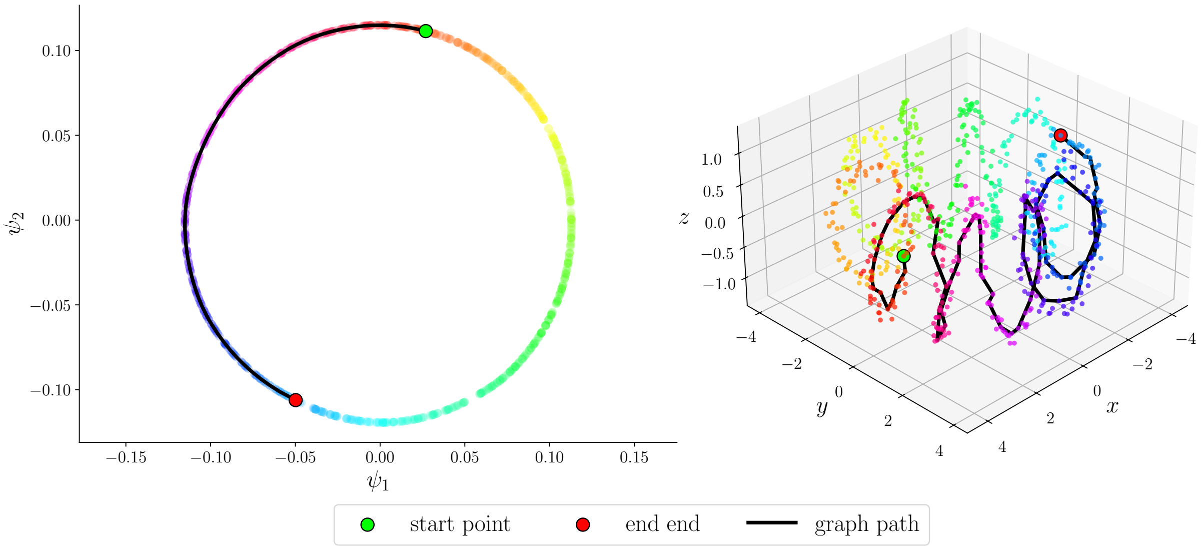


Figure 3: Left: graph path through diffusion coordinates. Right: graph path lifted from diffusion coordinates into ambient space. The graph path follows the underlying geometry. maybe better using two figures here with subcaptions.

We therefore construct kNN graphs from the diffusion coordinates as an approximate representation of our manifold I am not sure if this is complete. It is an approximation of the diffusion coordinate manifold, I think, but not the real manifold. Once we have a fully connected graph fully connected != complete right?, we can use a shortest path algorithm to find a path between any two nodes in our diffusion coordinates, which then will be made up of short paths corresponding to geodesics on the intrinsic dimenison. We therefore say that the shortest graph path represents a geodesic along the intrinsic data.

We demonstrate this in an example, continued from the previous section. SHOW GRAPH PATHS IN EXAMPLE DIFFUSION COORDINATE AND THE DISTANCE CORRELATION WITH GEODESIC FOR ALL PAIRS, NOT JUST CLOSE PAIRS. OR LIFTED POINTS IN THE GRAPH PATH INTO THE AMBIENT SPACE, SHOWING THE PATH FOLLOWS THE MANIFOLD.

Then we can see that graph paths through diffusion coordinates correspond to geodesics in the intrinsic geometry. Not sure how possible it is to make this claim. The toy model results look good, but idk if this is real

"The graph distance dG is a good approximation for the geodesic distance d in the original data and in any isometric embedding, as it will closely follow the actual manifold rather then cross in the ambient space." [24]

- Graph paths;
- ordinary graph-geodesic cost;
- the distinction between a Euclidean chord and a path constrained to observed neighbourhoods;
- Dijkstra's shortest-path problem.

2.5 Density-aware graph geometry??

Maybe worth going into extra detail here because this is novel

In this dissertation, we define a proxy for historical support as the density of diffusion coordinates. reference the section?. While the shortest graph paths represent the approximate shortest paths along the learned geometry of the data, we would like to traverse along the learned geometry such that the points we visit have high historical support.

To find the shortest paths, we use Dijkstra's algorithm, which includes the edge cost function

$$w_{ij} = \|x_i - x_j\|,$$

where the algorithm finds a path between nodes by minimising the sum of the edge costs. We intoroduce a modified edge cost which incorporates a sparsity penalty.

Suppose that ρ_i is the density of a node i in diffusion coordinates I want to say diffusion space, but I haven't defined that and I don't know if I will, then we can define a density potential as $V_i = -\log(\rho_i)$, which is high for low-density nodes and low for high-density nodes. Then we define the density-aware edge cost as

$$w_{ij}^{(\beta)} = \|x_i - x_j\| \exp\left(\beta \frac{V_i + V_j}{2}\right),$$

where β controls the strength of the sparsity avoidance. We can see that setting $\beta = 0$, we simply have the standard edge cost. However, increasing β means that the cost of an edge that involves a low-density node (high V_i) is now higher than it was previously. Depending on the density measure, it may be possible for points to have a density $\rho_i > 1$, which would correspond to a $V_i < 0$. This would then mean that an edge containing two high-density nodes (at least, both nodes have $rho > 1$), would have a smaller edge cost with the addition of the exponential term. This has the effect of making higher-density nodes more attractive to the algorithm.

Talk about why we use $V = -\log(\rho)$ instead of say $V = -\rho$. Implications etc, give intuition because it is used a lot later on, and it's not linear. Talk about how this cost effectively changes the shape of the graph which a standard graph algorithm uses to find the shortest path

- regions visited frequently by history correspond to high density
- low density means combinations that have not been seen before.
- show the negative correlation with distance and density.

3 Data and Preprocessing

3.1 Data sources and sample period

The analysis combines macroeconomic data from the Federal Reserve Economic Database (FRED) with U.S. industry portfolio returns from the Kenneth R. French Data Library. Following [3], the state vector is constructed jointly from macroeconomic variables and portfolio returns, allowing the learned geometry to reflect both macroeconomic conditions and associated variation across broad sectors of the financial market. The National Bureau of Economic Research (NBER) recession chronology is retained separately for subsequent comparison and is not included in the diffusion-map input.

Macroeconomic variables are taken from the May 2026 vintage of the FRED-MD database [21]. The original database contains 126 monthly U.S. macroeconomic series. After variable selection, 28 macroeconomic variables are retained for the analysis.

Monthly returns for the 10 industry portfolios are obtained from the Kenneth R. French Data Library [15]. The ten-industry classification provides broad coverage of U.S. equity-market performance while avoiding the introduction of a much larger number of closely related financial series. Since these observations are already expressed as monthly returns, no FRED-style differencing or growth-rate transformation is applied.

After aligning the two data sources, the initial sample runs from June 1963 to April 2026, giving 755 monthly observations. The transformations described in Section 3.4 reduce the usable sample by twelve months, giving a final sample from June 1964 to April 2026 with 743 observations.

NBER business-cycle peak and trough dates are used to construct a monthly recession indicator [22]. The indicator takes the value one during NBER-defined recession periods and zero otherwise.

3.2 Variable selection

Make sure it's obvious that this was not objective and could include bias. Also give the reasons you didn't include each variable in appendix

FRED-MD groups its macroeconomic series into broad categories representing different areas of the U.S. economy [21]. We use these groupings, together with the economic definitions of the individual series, to select a representative subset of variables. The aim is to retain broad coverage of macroeconomic conditions while avoiding the inclusion of several variables which describe essentially the same economic quantity.

Variable selection was based primarily on economic interpretation: the definitions of the FRED-MD series were inspected and closely related or derived variables were removed where they did not appear to provide sufficiently distinct economic information. This was a judgement-based preprocessing step rather than a formal statistical feature-selection procedure.

Reducing this redundancy may be particularly relevant for the diffusion-map construction. Each input variable contributes to the Euclidean distances used to form the affinity matrix, so including several closely related or directly derived variables could give one aspect of the economy disproportionate influence on the learned geometry.

This differs from the approach of [3], who retain all 124 variables in their specific FRED-MD dataset, so that stress scenarios can be specified using a broad description of the macroeconomic state rather than being restricted to a small number of preselected variables. This is well suited to their objective of conditioning on user-specified macroeconomic scenarios. The objective of the present dissertation is different: the primary object of interest is the geometry of the learned diffusion representation and the construction of paths through it. We therefore prioritise broad economic coverage while removing variables that are, by definition, closely related or directly derived from one another, in order to limit redundant weighting in the distance calculation.

The resulting dataset contains 28 macroeconomic variables spanning the major FRED-MD categories, together with the 10 industry portfolio return series described above. The complete set of retained variables and their transformations is reported in Appendix B, together with a correlation heatmap of the final 38-dimensional dataset.

3.3 Temporal alignment and missing observations

Both the FRED-MD and Kenneth R. French datasets are observed at a monthly frequency, so the two sources were aligned directly by month. Their common sample before applying the transformations described in Section 3.4 runs from June 1963 to April 2026, giving 755 monthly observations.

Only four values were missing from the aligned dataset, all occurring in the same month. Given the very small amount of missing data relative to the full sample, these values were filled using linear interpolation between the adjacent monthly observations.

The NBER recession indicator was aligned to the same monthly date index but retained separately from the variables used to construct the diffusion map.

3.4 Transformations

Maybe include a description of the transforms or just list what the options are. It feels a bit vague right now.

The FRED-MD series are transformed before constructing the diffusion map. This is important because many macroeconomic variables exhibit persistent trends in their levels. Without transformation, Euclidean distances between observations could therefore reflect long-run growth or differences in calendar time rather than differences in the prevailing economic state. We use the transformation codes provided with FRED-MD as the baseline specification, following [21].

Initial experiments using the recommended transformations showed that several activity variables produced exceptionally large observations around the COVID-19 disruption, particularly when represented using short-horizon differences or growth rates. Alternative transformations were therefore considered for a small subset of variables. For You've not introduced these variables, so it doesn't really make sense to have their names here.

$$\{\texttt{SRVPRD}, \texttt{PAYEMS}, \texttt{MANEMP}, \texttt{IPMANSICS}, \texttt{INDPRO}, \texttt{DPCERA3M086SBEA}, \texttt{CUMFNS}, \texttt{W875RX1}\},$$

we use year-on-year changes rather than the corresponding FRED-MD transformation. Year-on-year changes remain economically interpretable measures of activity while reducing the influence of exceptionally large movements occurring over a single month.

Two further modifications are made for labour-market variables. The unemployment rate, `UNRATE`, is retained in levels (this isn't obvious, maybe just say these are not changed, or define levels briefly) rather than first differences, and initial unemployment claims, `CLAIMSx`, are represented in logarithmic levels rather than log differences. These representations reduce the extreme tail behaviour generated by the COVID-19 observations while retaining economically meaningful information about the state of the labour market.

The subsequent quantile normalisation described in Section 3.5 further reduces the influence of extreme observations. Nevertheless, the choice of transformation remains relevant because month-on-month changes, year-on-year changes and levels represent different economic quantities and can lead to different joint configurations of the monthly observations. The alternative transformations are therefore retained as part of the final preprocessing specification, although they are not claimed to be uniquely optimal.

The use of year-on-year transformations requires twelve months of lagged observations. Consequently, the sample is reduced from 755 observations between June 1963 and April 2026 to 743 observations between June 1964 and April 2026. The transformation applied to each retained variable is reported in Appendix B.

3.5 Quantile normalisation

Find out how ties are broken

Following the transformations described above, the variables still differ in their marginal scales and tail behaviour. This is important for the diffusion-map construction used in this dissertation because affinities are calculated from Euclidean distances. Variables with greater numerical dispersion, or individual observations with exceptionally large values, may therefore have a disproportionate influence on the resulting geometry.

To reduce this sensitivity, a quantile transformation is applied independently to each variable. Each variable is transformed independently by replacing its observations according to their empirical rank and mapping these ranks approximately uniformly onto the interval [0,1]. Thus, an observation at the 10th percentile of a variable is mapped close to 0.1, the median is mapped close to 0.5, and an observation at the 90th percentile is mapped close to 0.9. The transformation therefore preserves the ordering of observations within each variable while removing differences in their original scales and strongly compressing extreme values.

A Gaussian target distribution was also considered during preliminary analysis. However, the leading diffusion coordinates remained dominated by observations associated with the COVID-19 disruption. The uniform target was therefore retained because its bounded support further limits the influence that any single extreme observation can have on the Euclidean distance calculation.

This robustness comes with an important trade-off. Quantile normalisation preserves the relative ordering of observations but not the original spacing between them. Consequently, an exceptionally severe shock is represented primarily by its position in the empirical distribution rather than by its magnitude in the original economic units. Distances in the resulting state space should therefore be interpreted as differences in the joint relative configuration of the variables, rather than as differences in their original magnitudes. The implications of this compression of extreme observations for stress testing are discussed further in Section 7.

3.6 Final analysis matrix

After variable selection, transformation, temporal alignment and quantile normalisation, the final analysis matrix is

$$\widetilde{Z} \in \mathbb{R}^{743 \times 38},$$

where each row represents one monthly observation from June 1964 to April 2026 and the columns contain 28 macroeconomic variables and 10 industry portfolio returns. The NBER recession indicator is retained separately and is not included in the diffusion-map construction.

4 Numerical methods

4.1 Overview of the computational pipeline

This is where you give a clear picture of what you actually are going to do. Reference the different sections, etc.

raw data $\rightarrow$ transformation $\rightarrow$ diffusion map $\rightarrow$ density and graph $\rightarrow$ stress paths $\rightarrow$ lifting

? $\rightarrow$ HMM regime analysis

In this section, we describe the entire computational pipeline. Firstly, using our transformed and standardised data, we find a diffusion map, which captures the potential non-linear structure of the macroeconomic manifold on which the data lies. Using that diffusion map, we choose an appropriate embedding dimension, and now we have our new coordinate system. Using these new coordinates, we construct a k-nearest-neighbours graph and compute an estimation of the densities. We then implement our density-aware graph path model to interpolate a path between a variety of specifically chosen economic endpoints. We compare this path to various other types of paths, specifically, a linear path in diffusion space, and a standard graph path. Then we can lift these paths back to the macro space, and there we can compare different paths in real coordinates to see how well our model does.

I'm not sure if I should include the above section yet, so I haven't worked on it. You can skip this part and start with Section 4.2.

4.2 Diffusion-map construction

The diffusion-map construction introduced in Section 2.3 was applied to the transformed macroeconomic state matrix

$$\widetilde{Z} = \begin{pmatrix} \widetilde{z}_1^\top \\ \vdots \\ \widetilde{z}_N^\top \end{pmatrix} \in \mathbb{R}^{N \times D},$$

where each row $\widetilde{z}_i \in \mathbb{R}^D$ represents one monthly macroeconomic observation and each column corresponds to one of the variables described in Section 3. The NBER recession indicator was retained for subsequent external comparison, but was not included as an input to the embedding.

The construction follows the diffusion-map framework from Section 2.3, but is restated here in computational form to make explicit how the empirical embedding was obtained. Pairwise squared Euclidean distances were first calculated as

$$\delta_{ij} = \|\widetilde{z}_i - \widetilde{z}_j\|^2, \qquad i, j = 1, \ldots, N.$$

These distances were converted into Gaussian affinities,

$$W_{ij} = \exp\left(-\frac{\delta_{ij}}{\varepsilon}\right),$$

where $\varepsilon > 0$ controls the neighbourhood scale represented by the kernel.

The kernel degrees

$$q_i = \sum_{j=1}^{N} W_{ij}$$

were then used to construct the density-normalised affinity matrix

$$W_{ij}^{(\alpha)} = \frac{W_{ij}}{q_i^\alpha q_j^\alpha}.$$

James asked why we don't take $\sqrt{}$ here Throughout the main empirical analysis, the density-normalisation parameter was fixed at

$$\alpha = 1.$$

As discussed in Section 2.3, this choice removes the leading influence of non-uniform sampling density from the diffusion operator and is therefore appropriate when the objective is to recover the geometry represented by the observations rather than to preserve their empirical sampling frequency.

The row sums of the normalised affinity matrix are

$$d_i^{(\alpha)} = \sum_{j=1}^{N} W_{ij}^{(\alpha)},$$

and define the diagonal matrix

$$D_\alpha = \operatorname{diag}\left(d_1^{(\alpha)}, \ldots, d_N^{(\alpha)}\right).$$

The corresponding diffusion transition matrix is

$$P = D_\alpha^{-1} W^{(\alpha)}.$$

why? This is just by definition, I think; doing this means that the row sums are equal to 1, so then each element corresponds to the probability as desired. This is the Markov operator whose right eigenvectors define the diffusion coordinates introduced in Section 2.3.

The highlighted red pattern is seen a lot and James says it's an LLM tell. Don't use this In the numerical implementation, however, the eigenpairs were not obtained by directly diagonalising the generally non-symmetric matrix P. Instead, the symmetric matrix

$$S = D_\alpha^{-1/2} W^{(\alpha)} D_\alpha^{-1/2}$$

was constructed. Since

$$P = D_\alpha^{-1/2} S D_\alpha^{1/2},$$

the matrices P and S are similar and therefore have the same eigenvalues. The symmetry of S allows its eigenpairs

$$S u_k = \lambda_k u_k$$

to be computed using a symmetric eigensolver. The corresponding right eigenvectors of the diffusion transition matrix are then recovered through

$$\phi_k = D_\alpha^{-1/2} u_k.$$

Indeed,

$$P\phi_k = D_\alpha^{-1/2} S u_k = \lambda_k \phi_k,$$

so the recovered ϕ_k are precisely the eigenvectors entering the diffusion-map representation.

The eigenpairs were ordered by decreasing eigenvalue,

$$1 = \lambda_0 \geq \lambda_1 \geq \lambda_2 \geq \cdots,$$

and the trivial eigenpair (λ_0, ϕ_0) What does this pair represent? This is the smoothest way to assign coordinates, but it gives no information. Can maybe talk about because we normalised P, then we have that the biggest eigenvalue corresponds to the constant eigenvector... or choose a response which talks about whether this represents an intercept? Idk if it does though was removed. At diffusion time $t = 1$, retaining the first r non-trivial modes gives

$$\Psi(\tilde{z}_i) = (\lambda_1 \phi_1(i), \ldots, \lambda_r \phi_r(i)) \in \mathbb{R}^r.$$

James was asking whether this was PCA in D_α inner product - not really sure. I think that diffusion maps are a special case of something called "kernel PCA" (not sure if that's the name) - "all kernel-based manifold learning methods are special cases of kernel PCA [10]" Equivalently, collecting the coordinates for all observations gives the embedding matrix

$$\Psi = \begin{bmatrix} \lambda_1 \phi_1 & \cdots & \lambda_r \phi_r \end{bmatrix} \in \mathbb{R}^{N \times r}.$$

The complete computational procedure is summarised in Algorithm 1.

Algorithm 1 Diffusion-map embedding

Input: $\widetilde{Z} \in \mathbb{R}^{N \times D}$, ε, α, r
Output: Embedding coordinates $\Psi \in \mathbb{R}^{N \times r}$
$\delta_{ij} \leftarrow \|\widetilde{z}_i - \widetilde{z}_j\|^2$
$W_{ij} \leftarrow \exp(-\delta_{ij}/\varepsilon)$
$q_i \leftarrow \sum_{j=1}^{N} W_{ij}$
$W_{ij}^{(\alpha)} \leftarrow W_{ij}/(q_i^\alpha q_j^\alpha)$
$d_i^{(\alpha)} \leftarrow \sum_{j=1}^{N} W_{ij}^{(\alpha)}$
$D_\alpha \leftarrow \operatorname{diag}\left(d_1^{(\alpha)}, \ldots, d_N^{(\alpha)}\right)$
$P \leftarrow D_\alpha^{-1} W^{(\alpha)}$
$S \leftarrow D_\alpha^{-1/2} W^{(\alpha)} D_\alpha^{-1/2}$
Compute $S u_k = \lambda_k u_k$
$\phi_k \leftarrow D_\alpha^{-1/2} u_k$
Sort (λ_k, ϕ_k) in descending order of λ_k
Discard the trivial eigenpair (λ_0, ϕ_0)
$\Psi \leftarrow [\lambda_1 \phi_1, \ldots, \lambda_r \phi_r]$
return Ψ

Algorithm 1 requires two choices that materially determine the resulting representation: the kernel bandwidth ε, which controls the scale at which local observations are connected, and the retained dimension r, which determines the number of non-trivial diffusion modes used to represent the data. Rather than fixing these quantities a priori, both were assessed empirically using the diagnostics described in the following sections. The selected values were then held fixed for the principal graph construction, density estimation and stress-path analysis, with sensitivity to these choices examined separately in Section 5.5.

4.3 Diffusion parameter selection

The kernel bandwidth ε controls the locality of the Gaussian affinity kernel and was selected before fixing the embedding dimension. As an initial scale diagnostic, the kernel mass

$$T(\varepsilon) = \sum_{i,j} \exp\left(-\frac{\|\widetilde{z}_i - \widetilde{z}_j\|^2}{\varepsilon}\right)$$

was evaluated over a logarithmically spaced range of bandwidths, together with its local log–log slope, as described in Section 2.3. This diagnostic identifies scales at which the total kernel mass changes most rapidly how?. However, in a finite and non-uniformly sampled dataset, the bandwidth selected from the slope alone may produce a poorly connected graph why? This might be something worth discussing; a mini result. I may also discuss all of this in the background bit. The kernel-mass rule was therefore used to identify a plausible bandwidth range rather than as an automatic selection criterion.

Candidate bandwidths were subsequently compared using the connectivity and spectral properties of the resulting diffusion operator. Let

$$q_i(\varepsilon) = \sum_j W_{ij}(\varepsilon)$$

denote the unnormalised kernel degree of observation i. The minimum degree,

$$q_{\min}(\varepsilon) = \min_i q_i(\varepsilon),$$

was used to detect observations that were effectively isolated from the remainder of the sample. Since $W_{ii} = 1$, a value of q_i close to one indicates that almost all of the degree is contributed by the self-affinity. The heterogeneity of the degree distribution was measured using

$$\mathrm{CV}_q(\varepsilon) = \frac{\mathrm{sd}\{q_i(\varepsilon)\}}{\mathrm{mean}\{q_i(\varepsilon)\}}.$$

A large coefficient of variation indicates that the graph connectivity is concentrated unevenly across observations, while comparison across neighbouring bandwidths provides a check on the stability of the connectivity structure.

The leading non-trivial eigenvalues of the transition matrix P were also compared across candidate bandwidths. Particular attention was given to the stability of the dominant eigenvalues and the gaps between consecutive eigenvalues under small changes in ε. The corresponding diffusion coordinates were inspected to exclude bandwidths for which the leading eigenvectors were nearly constant over most observations but contained isolated sharp spikes, as this behaviour is indicative of weakly connected or nearly isolated graph components.

Combining these diagnostics, the bandwidth was fixed at

$$\varepsilon = 3.$$

This choice provided adequate graph connectivity while retaining sufficient locality in the kernel. The kernel-mass diagnostic, degree statistics and spectral comparisons supporting this choice are reported in Appendix C.

The following paragraph was unclear. Why not use the full spectrum? Explain the compromises. We want to reduce the dimensions to make things simpler and reduce noise maybe. The embedding dimension was not selected from the eigenspectrum alone. Candidate dimensions were assessed using three complementary criteria: the leading non-trivial eigenspectrum, the separation of near-collision pairs and the reconstruction performance of the local-neighbour lifting procedure, outlined in Section 4.7.

The eigenspectrum was used to identify the number of dominant diffusion modes and to locate large gaps between consecutive non-trivial eigenvalues. Since a spectral gap does not by itself guarantee that the retained coordinates distinguish all economically different observations, an additional near-collision diagnostic was introduced. For an embedding of dimension r, pairs of observations separated by no fewer than twelve months were ranked according to their Euclidean distance in diffusion space. The closest 1% of eligible pairs were defined as near-collisions. Their distance ranks were then recomputed after adding coordinate $r + 1$. The proportion of these pairs that left the closest 1%, 5% or 10% in the higher-dimensional embedding measured how frequently the additional coordinate separated observations that had appeared nearly identical in the lower-dimensional representation. A large exit proportion therefore indicates that coordinate $r + 1$ contains substantial information not represented by the first r coordinates.

The ability of each candidate embedding to reconstruct the transformed macroeconomic observations was assessed through the local-neighbourhood lift. If $\widehat{Z}^{(r)}$ denotes the observations reconstructed from an r-dimensional embedding, reconstruction accuracy was summarised using the root mean squared error and the proportion of variance recovered,

$$R^2_{\text{lift}}(r) = 1 - \frac{\|Z - \widehat{Z}^{(r)}\|_F^2}{\|Z - \overline{Z}\|_F^2},$$

why do you use this? What does it mean? where $\overline{Z}$ contains the variable-wise sample means. This diagnostic measures the extent to which the retained diffusion coordinates preserve information required to recover the original transformed variables.

Finally, low-dimensional projections were inspected after colouring observations by calendar time, recession status and selected macroeconomic variables. These visualisations were used as a qualitative check that the retained coordinates represented distinct and economically interpretable directions of variation, rather than as a standalone dimension-selection rule.

The combined diagnostics led to the selection

$$r = 4.$$

The fourth coordinate provided a material improvement in near-collision separation and lifted reconstruction relative to the three-dimensional embedding, whereas the incremental benefit of adding a fifth coordinate was comparatively small. The numerical comparisons and embedding visualisations are presented in Section 5.1.

4.4 Graph construction and density estimation

Following the construction of the diffusion embedding, an undirected k-nearest-neighbours graph was formed from the observed diffusion coordinates,

$$\mathcal{G} = (\mathcal{V}, \mathcal{E}), \qquad \mathcal{V} = \{1, \dots, N\}.$$

A directed edge from node i to node j was first introduced whenever ψ_j was among the k nearest neighbours of ψ_i in diffusion space what is diffusion space? Make this clear. The graph was then symmetrised by taking the union of the directed edge sets, so that $(i, j) \in \mathcal{E}$ whenever either observation selected the other as a neighbour. Consequently, the average degree of the resulting graph may be slightly greater than k. The Euclidean length assigned to each edge was

$$\ell_{ij} = \|\psi_i - \psi_j\|.$$

The nearest-neighbour search was implemented using Scikit-Learn's `NearestNeighbors` class.

Candidate values of k were assessed using several complementary graph diagnostics. First, the graph was required to be connected, ensuring that a path existed between any selected pair of endpoints. All candidate values considered satisfied this condition. Second, the graph density

$$d_{\mathcal{G}} = \frac{2|\mathcal{E}|}{N(N-1)}$$

was monitored to ensure that the graph remained local and did not approach a complete graph as k increased.

The presence of unusually long edges within shortest paths was also examined. A systematic subsample containing every fifth node was formed, and shortest paths were calculated between every unordered pair of nodes in this subsample. For each path Γ, the statistic

$$R_{\text{edge}}(\Gamma) = \frac{\max_{(i,j)\in\Gamma} \ell_{ij}}{\operatorname{median}_{(i,j)\in\Gamma} \ell_{ij}}$$

was recorded. A large value indicates that the path depends upon an edge whose length is substantially greater than its typical edge length. Such an edge may act as an artificial shortcut across a sparsely sampled region, causing the graph path to represent a non-local jump rather than gradual movement through the sampled manifold why is this bad? Is this not a part of the geometry? Why don't we want this? This should be made clear in the small example. The distribution of this statistic was therefore compared across candidate values of k.

A final value of

$$k = 15$$

was selected. Surrounding values in the searched grid yielded similar metrics, so would be suitable candidates. This choice of k was somewhat an arbitrary one being the middle value in our grid. Further analysis could be done in order to calibrate this choice of k with graph path jump lengths to match to historically observed jumps. Is this what I should say, or is it OK to just say it was arbitrary and move on?

A relative density score was then estimated from the observations in diffusion space. For an arbitrary point $\psi \in \mathbb{R}^r$, define

$$\widehat{\rho}(\psi; h) = \sum_{j=1}^{N} \exp\left(-\frac{\|\psi - \psi_j\|^2}{h}\right),$$

where $h > 0$ is another bandwidth parameter which controls the locality of the density estimate. At an observed node, this gives

$$\widehat{\rho}_i = \widehat{\rho}(\psi_i; h).$$

Since only relative density is required for the path construction, the kernel score was normalised by its mean over the observed nodes,

$$\overline{\rho} = \frac{1}{N}\sum_{i=1}^{N} \widehat{\rho}_i, \qquad \rho(\psi) = \frac{\widehat{\rho}(\psi; h)}{\overline{\rho}},$$

with $\rho_i = \rho(\psi_i)$.

Candidate values of h were compared according to whether the resulting density field distinguished well-supported and sparsely supported observations without becoming nearly constant or becoming almost binary (choosing a very small value of h results in only a few values of high-density, whereas most points are assigned a similar level of low density, hence almost binary). The value of h we chose means that the resulting density of points varies smoothly. Is this a good enough explanation? The value which fit these conditions was one half of the value of the median squared distance between all diffusion coordinates.

For numerical stability, a small density floor $\delta = 1 \times 10^{-6}$ was imposed,

$$\rho_{i,\delta} = \max\{\rho_i, \delta\},$$

however, for our data and final choice of h, this floor was never actually imposed, as all the density scores were above the value of δ. If it was imposed, then this would impact the normalisation, right?

We define a density potential at node i as

$$V_i = -\log(\rho_{i,\delta}).$$

Under this construction, observations surrounded by many nearby historical states have relatively low potential, whereas isolated observations have relatively high potential.

The density score is interpreted as a measure of *historical support under the geometry learned by the diffusion map*. It does not by itself establish that a state is economically plausible or dynamically attainable. These statements are backed up by results in Section 5.2, where we evaluate the diffusion density's empirical relationship with recognised periods of economic stress and with proximity to historical observations in the transformed macroeconomic space is examined in the results.

4.5 Stress-path construction

4.5.1 Endpoint selection

Four historically motivated start–stress endpoint pairs were considered. Both the start and stress endpoints were required to correspond to observed dates and hence to existing nodes of the diffusion graph why? So we can test the method directly instead of having to deal with out-of-sample methods. Can compare actual paths too?. No out-of-sample extension was therefore required for the principal stress-path analysis.

The endpoints were selected to represent economically interpretable transitions from comparatively benign macroeconomic conditions to distinct forms of historical stress how? how were these selected? Looked at the individual macro variables and checked dates with news. The specific dates, scenario labels and summary economic characteristics are introduced in the empirical results. More detailed historical discussion is provided separately where required.

4.5.2 Path definitions

Three paths were constructed between each pair of endpoints: a linear path through diffusion space, an ordinary graph shortest path and a density-aware graph shortest path. Comparing these constructions separates the effect of restricting movement to the empirical graph from the additional effect of favouring historically well-supported regions.

Let i_{start} and i_{stress} denote the graph nodes corresponding to the selected endpoints. A graph path is represented by an ordered sequence of node indices

$$\Gamma = (i_1, \dots, i_L),$$

where

$$i_1 = i_{\text{start}}, \qquad i_L = i_{\text{stress}}, \qquad (i_m, i_{m+1}) \in \mathcal{E}.$$

For a specified set of positive edge costs w_{ij}, the shortest path solves

$$\Gamma^\star = \underset{\Gamma : i_{\text{start}} \to i_{\text{stress}}}{\arg\min} \sum_{m=1}^{L-1} w_{i_m i_{m+1}}.$$

The optimisation was implemented using Dijkstra's Algorithm via SciPy's graph shortest-path routine.

The ordinary graph path assigns each edge its Euclidean diffusion-space length,

$$w_{ij}^{(0)} = \ell_{ij} = \|\psi_i - \psi_j\|.$$

This path is constrained to move through observed graph nodes but does not explicitly distinguish between densely and sparsely supported regions.

To incorporate historical support, the density-aware edge cost was defined for $\beta \geq 0$ by

$$w_{ij}^{(\beta)} = \|\psi_i - \psi_j\| \exp\left(\beta \frac{V_i + V_j}{2}\right).$$

The average of the endpoint potentials approximates the density potential encountered while traversing the edge. Since low-density nodes have larger values of V_i, edges passing through sparsely supported regions receive a higher cost. The corresponding density-aware path is

$$\Gamma_\beta = \underset{\Gamma : i_{\text{start}} \to i_{\text{stress}}}{\arg\min} \sum_{(i,j) \in \Gamma} w_{ij}^{(\beta)}.$$

When $\beta = 0$, this reduces to the ordinary graph shortest path. Increasing β places progressively greater emphasis on historical support relative to direct path length.

Candidate values of β were compared according to the trade-off between improved density support and the additional length or detour introduced into the path. Stability across nearby values was also examined. The retained specification is used for the principal analysis, while the sensitivity of the results to β is considered in the robustness analysis.

As an unconstrained benchmark, a straight line was constructed between the same endpoints in diffusion space,

$$\gamma_{\text{lin}}(s) = (1-s)\psi_{\text{start}} + s\psi_{\text{stress}}, \qquad s \in [0,1].$$

this is the LLM pattern again highlighted in red The path was evaluated on an equally spaced grid of values of s. Unlike the graph paths, its intermediate points need not coincide with observed historical states and it may pass through regions containing little empirical support. It therefore provides a diffusion coordinate space linear benchmark against which the effects of graph restriction and density weighting can be assessed This seems like a very weak baseline. You need to describe what the diffusion space is better for this ever to make sense.

The graph and linear paths are compared using the path-length, density-support and macroeconomic evaluation measures defined in Section 4.6.

4.6 Stress-path evaluation

The stress paths are evaluated according to three complementary properties: the historical support encountered along the path, the distance travelled through diffusion space, and the regularity of the resulting trajectory. These quantities are considered jointly, since a path may avoid sparsely supported regions only by taking a substantially longer or more irregular route.

For evaluation, let a discretised path be written as

$$\gamma = (\gamma_0, \ldots, \gamma_L),$$

where $\gamma_\ell \in \mathbb{R}^r$. For a graph path, the points γ_ℓ are observed diffusion coordinates ψ_{i_ℓ}, whereas for the linear benchmark they are interpolated points along the straight line defined in Section 4.5.

The principal evaluation criterion is the density encountered along the path. For an interpolated point γ_ℓ that does not coincide with an observed state, the density score is evaluated against the historical diffusion coordinates using the same kernel and bandwidth as in Section 4.4,

$$\widehat{\rho}(\gamma_\ell) = \sum_{j=1}^{N} \exp\left(-\frac{\|\gamma_\ell - \psi_j\|^2}{h}\right).$$

This is normalised using the same full-sample scale

$$\overline{\rho} = \frac{1}{N}\sum_{i=1}^{N} \widehat{\rho}(\psi_i),$$

giving

$$\rho(\gamma_\ell) = \frac{\widehat{\rho}(\gamma_\ell)}{\overline{\rho}}.$$

A leave-one-out modification is used whenever the evaluated point corresponds to an observed graph node. For $\gamma_\ell = \psi_i$, its evaluation density is

$$\widehat{\rho}_{-i}(\psi_i) = \sum_{\substack{j=1 \\ j\neq i}}^{N} \exp\left(-\frac{\|\psi_i - \psi_j\|^2}{h}\right), \qquad \rho_{-i}(\psi_i) = \frac{\widehat{\rho}_{-i}(\psi_i)}{\overline{\rho}}.$$

James warns that this may be unstable - what if there is a point within some $\delta \ll 1$? Maybe explain/fix, but you can also just check that no such points exist in the data. I am not currently sure how it is unstable. Removing the observation's self-contribution prevents graph nodes from receiving an automatic affinity of one purely because they belong to the historical sample. This leave-one-out adjustment is used only for path evaluation; the density field used to construct the density-aware graph remains that defined in Section 4.4. The same convention is applied to any observed endpoint appearing on the linear path. After evaluation, the density floor δ is applied as before.

Denote the resulting evaluation density at γ_ℓ by $\rho_{\text{eval}}(\gamma_\ell)$. The primary support metric is the minimum density,

$$M_{\min}(\gamma) = \min_{0 \le \ell \le L} \rho_{\text{eval}}(\gamma_\ell),$$

which identifies the least historically supported state encountered along the path. Since this metric is determined by a single point, the overall level of support is also summarised by the path-average density. To account for differences in path discretisation, this is weighted by the diffusion-space length of each segment,

$$M_{\text{mean}}(\gamma) = \frac{1}{L_{\text{diff}}(\gamma)} \sum_{\ell=0}^{L-1} \frac{\rho_{\text{eval}}(\gamma_\ell) + \rho_{\text{eval}}(\gamma_{\ell+1})}{2} \|\gamma_{\ell+1} - \gamma_\ell\|,$$

where

$$L_{\text{diff}}(\gamma) = \sum_{\ell=0}^{L-1} \|\gamma_{\ell+1} - \gamma_\ell\|$$

is the total path length in diffusion coordinate space. This provides a measure of the additional distance required for a path to remain in better-supported regions of the learned diffusion geometry.

Finally, path regularity is measured through changes in direction between successive segments. Define the unit direction vector

$$u_\ell = \frac{\gamma_{\ell+1} - \gamma_\ell}{\|\gamma_{\ell+1} - \gamma_\ell\|}, \qquad \ell = 0, \dots, L-1.$$

The turning angle at an interior path point is

$$\theta_\ell = \arccos\left(u_{\ell-1}^\top u_\ell\right), \qquad \ell = 1, \dots, L-1,$$

and the total direction change is

$$T_{\text{turn}}(\gamma) = \sum_{\ell=1}^{L-1} \theta_\ell.$$

A straight path has zero direction change, while larger values indicate a more tortuous trajectory through diffusion space. This seems inconsistent, is the euclidian inner product meaningful? I just used it as a visual aid, because the 2d projections of the path may look straight even though they are very jagged in other dimensions. Again, you mention diffusion space and being 'tortuous' in it. Change this wording for one, and check that even makes sense.

LLM pattern again The path constructions are therefore compared using minimum and average historical support together with path length in diffusion coordinate space and mean direction change. The objective is not to optimise any one of these quantities independently, but to assess whether the graph-based constructions achieve greater historical support without requiring excessive detours or irregularity. The empirical comparisons are reported in Section 5.3.

4.7 Lifting paths into macroeconomic space

The stress paths constructed in diffusion space must ultimately be expressed in terms of the transformed macroeconomic variables. For graph paths, no estimated inverse mapping is required. Each graph vertex corresponds to an observed diffusion coordinate ψ_i and therefore to its associated macroeconomic observation $\tilde{z}_i$. Hence, for a graph path

$$\Gamma = (i_1, \dots, i_L),$$

the corresponding macroeconomic path is obtained directly as

$$(\tilde{z}_{i_1}, \ldots, \tilde{z}_{i_L}).$$

In contrast, the intermediate points of the linear benchmark generally do not correspond to observed states and must therefore be lifted from diffusion space back to macroeconomic space.

Two lifting procedures were considered. The first is a global linear inverse map, following the approach of Baker et al. [3]. Let

$$\psi_i^{(c)} = \psi_i - \overline{\psi}, \qquad \tilde{z}_i^{(c)} = \tilde{z}_i - \overline{z},$$

denote the centred diffusion and macroeconomic coordinates. The lifting matrix is obtained from

$$H = \arg\min_{B \in \mathbb{R}^{D \times r}} \sum_{i=1}^{N} \left\| \tilde{z}_i^{(c)} - B\psi_i^{(c)} \right\|^2,$$

Is this well posed? (it looks like it isn't) and an arbitrary diffusion-space point ψ is lifted according to

$$\hat{z}_{\text{lin}}(\psi) = H(\psi - \overline{\psi}) + \overline{z}.$$

Centring the variables before fitting and subsequently restoring $\overline{z}$ is equivalent to allowing the linear reconstruction to contain an intercept. This construction will be referred to as the Baker-style lift.

The second procedure is a local-neighbourhood lift (LNL). For a point ψ, let $\mathcal{N}_m(\psi)$ denote its m nearest historical observations in diffusion space. The lifted state is defined as the weighted average

$$\hat{z}_{\text{LNL}}(\psi) = \sum_{j \in \mathcal{N}_m(\psi)} a_j(\psi)\tilde{z}_j,$$

where

$$a_j(\psi) = \frac{\exp\left(-\|\psi - \psi_j\|^2/\tau\right)}{\sum_{k \in \mathcal{N}_m(\psi)} \exp\left(-\|\psi - \psi_k\|^2/\tau\right)}.$$

justify The weights are non-negative and satisfy

$$\sum_{j \in \mathcal{N}_m(\psi)} a_j(\psi) = 1.$$

Unlike the global linear lift, this construction allows the inverse approximation to vary across different regions of the embedding.

A grid search was performed to find the scale parameter τ and the neighbourhood size m. The parameter pair was chosen to minimise the leave-one-out reconstruction error. For our diffusion coordinates, we chose

$$\tau = 0.15 D_{\text{med}}^2, \quad m = 30,$$

where D_{med}^2 is the median squared distance between the diffusion coordinates $\Psi(i)$.

Since the endpoints of each linear stress path correspond to known historical observations, their macroeconomic values are available directly. The lifted linear paths were therefore anchored to the exact observed start and stress states rather than using the reconstructed values produced by either lifting method at these two points.

To assess the ability of each lifting procedure to recover historical observations, leave-one-out reconstruction diagnostics were calculated. For each observation i, the corresponding macroeconomic state was removed from the training sample and the lifting procedure was constructed using the remaining $N-1$ observations. The held-out diffusion coordinate ψ_i was then lifted to obtain $\hat{z}_{-i}$ and compared with the true state $\tilde{z}_i$. Reconstruction performance was summarised using the root mean squared error

$$\mathrm{RMSE} = \sqrt{\frac{1}{ND}\sum_{i=1}^{N} \|\widetilde{z}_i - \widehat{z}_{-i}\|^2},$$

and the proportion of variance recovered,

$$R^2_{\text{lift}} = 1 - \frac{\sum_{i=1}^{N} \|\widetilde{z}_i - \widehat{z}_{-i}\|^2}{\sum_{i=1}^{N} \|\widetilde{z}_i - \overline{z}\|^2}.$$

These quantities were compared with a baseline reconstruction that assigns each held-out observation the mean of the remaining training observations this is a very weak baseline, what else could you use? I now don't include the baseline score, but this is a part of the R^2 calculation, so you may want to explain it. As the leave-one-out error is also used in selecting the neighbourhood size for LNL, these quantities are treated as cross-validated reconstruction diagnostics rather than as an entirely independent test set.

A second diagnostic was used to detect whether either lifting procedure mapped the interpolated linear paths unusually far from the historical macroeconomic cloud. For an interior lifted path point $\widehat{z}_\ell$, define

$$d_{\mathrm{NN}}(\widehat{z}_\ell) = \min_{1\leq j\leq N} \|\widehat{z}_\ell - \widetilde{z}_j\|.$$

The median and 95th percentile of these distances were calculated along each lifted linear path. The fixed historical endpoints were excluded from this diagnostic, since their nearest-neighbour distance is zero by construction. Still might have endpoint effects? I think this is talking about if there is a point very close to the endpoint. Or it might be talking about the fact that a 95th percentile doesn't tell us about extremes

To provide a reference scale, these path distances were compared with leave-one-out nearest-neighbour distances among the historical observations,

$$d_{\mathrm{NN},-i}(\widetilde{z}_i) = \min_{j\neq i} \|\widetilde{z}_i - \widetilde{z}_j\|.$$

The corresponding historical median and 95th percentile indicate the typical and upper-tail separation between observed macroeconomic states. The lifted-path diagnostic therefore tests whether the inverse mapping produces states lying unusually far from the empirical macroeconomic cloud. It is not interpreted as a direct measure of economic plausibility.

The lifting procedure used in the principal stress-path analysis was selected by considering both cross-validated reconstruction accuracy and the extent of extrapolation observed along the lifted linear paths. The numerical comparison between the Baker-style and local-neighbourhood lifts is presented in Section 5.4.

4.8 Robustness design

In this section, we describe the robustness checks we performed to ensure the validity of our results. We hope don't say this. We just test to see that our qualitative result is stable: density-aware graph paths avoid low-density regions more effectively than straight-line interpolation. Additionally, we want again, don't say this. This is a bias to see a stable difference in the path diagnostics described in Section 4.6 between an ordinary and a density-aware graph path.

4.8.1 Sensitivity to core path parameters

The stress-path construction depends on several tuning parameters introduced in the preceding sections. Although the baseline values are selected using the diagnostics described earlier, it is important to establish that the principal conclusions are not specific to a single parameter configuration. Sensitivity analysis was therefore performed for the diffusion-map bandwidth ε, the number of nearest neighbours k used to construct the graph, and the density-penalty parameter β. In each experiment, one parameter was varied while the remaining parameters were held at their baseline values,

$$\varepsilon = 3, \qquad k = 15, \qquad \beta = 1.5.$$

The analysis was repeated for each of the four start–stress event pairs.

The bandwidth was varied over

$$\varepsilon \in \{1, 1.5, 2, 2.5, 3, 3.5, 4, 4.5, 5\}.$$

For each value of ε, the diffusion representation was reconstructed and the subsequent density estimation, graph construction and path calculation were repeated using the baseline values of k and β. The purpose of this experiment differs from the bandwidth diagnostics used to select ε in Section 4.3. Those diagnostics assess whether the diffusion operator itself is sufficiently well behaved, whereas the present analysis asks whether the resulting stress-path conclusions remain stable over a reasonable neighbourhood of the selected bandwidth.

Graph sensitivity was assessed using

$$k \in \{5, 10, 15, 20, 25, 30\}.$$

For each value of k, the symmetric nearest-neighbour graph was reconstructed on the fixed baseline diffusion embedding, after which both the ordinary and density-aware shortest paths were recalculated. This tests whether the conclusions depend strongly on the particular level of graph locality used to approximate movement through the sampled manifold.

For both ε and k, robustness was assessed by comparing the density-aware path directly with the corresponding ordinary graph path under the same parameter configuration. Let $M_{\min}$, M_{mean}, L_{diff} and D_{turn} denote respectively the minimum path density, arc-length-weighted mean density, latent path length and total turning angle defined in Section 4.6. For a generic metric M, define

$$\Delta M = M^{(\text{aware})} - M^{(\text{ordinary})}.$$

The four reported differences are therefore

$$\Delta M_{\min}, \qquad \Delta M_{\text{mean}}, \qquad \Delta L_{\text{diff}}, \qquad \Delta D_{\text{turn}}.$$

Positive values of the first two quantities indicate that the density-aware path travels through regions of greater historical support than the ordinary graph path. By contrast, positive values of ΔL_{diff} and ΔD_{turn} indicate the geometric cost associated with this change, in the form of a longer or more strongly turning route. Expressing the results as differences isolates the effect of density awareness from changes to the underlying embedding or graph that affect both path types simultaneously.

The density penalty was varied over

$$\beta \in \{0, 0.25, 0.5, 0.75, 1, 1.5, 2\}.$$

Here the diffusion embedding, density field and k-nearest-neighbour graph were held fixed at their baseline specifications, and only the edge-cost penalty was changed. Since $\beta = 0$ reduces the density-aware objective to the ordinary graph shortest-path problem, a separate ordinary comparison is unnecessary in this experiment. Instead, the absolute values of

$$M_{\min}, \qquad M_{\text{mean}}, \qquad L_{\text{diff}}, \qquad D_{\text{turn}}$$

were followed as β increased. This provides a direct view of the trade-off between avoiding regions of weak historical support and accepting additional path length or curvature.

No smooth dependence on these parameters is assumed. The stress paths are solutions to a discrete shortest-path problem, and changes in the embedding, graph connectivity or edge weights may cause the optimal route to switch between different sequences of observed states. Robustness is therefore assessed primarily through the persistence of the substantive comparison between density-aware and ordinary paths across the tested parameter ranges, rather than through smooth variation of the individual metrics. The corresponding sensitivity results are reported in Section 5.5.

4.8.2 Sensitivity to density estimation

The density-aware path construction depends on the estimate of historical support in diffusion space. We therefore test whether the principal empirical conclusion (that density-aware graph paths avoid regions of weak historical support relative to the linear and ordinary graph benchmarks) is robust to alternative definitions of the density field. In this experiment, the diffusion embedding, nearest-neighbour graph and density-penalty parameter are held fixed at their baseline specifications, while the density estimator used to construct the potential is varied.

The baseline estimator used throughout the main analysis is the graph-degree density introduced in Section 4.4. For a general point $x \in \mathbb{R}^r$ in diffusion space, its unnormalised density estimate is

$$\widehat{\rho}_G(x) = \sum_{j=1}^{N} \exp\left(-\frac{\|x - \psi_j\|^2}{h_{\text{dens}}}\right).$$

As previously, this is rescaled by the mean density at the observed diffusion coordinates,

$$\rho_G(x) = \frac{\widehat{\rho}_G(x)}{N^{-1}\sum_{i=1}^{N} \widehat{\rho}_G(\psi_i)}.$$

To assess sensitivity to this choice, we also consider kernel density estimators (KDEs). For a radial kernel K and bandwidth $b > 0$, the KDE at a general diffusion-space point x takes the form

$$\widehat{\rho}_K(x) = \frac{1}{Nb^r} \sum_{j=1}^{N} K\left(\frac{\|x - \psi_j\|}{b}\right).$$

Three kernel forms are considered. The Gaussian kernel is

$$K_G(u) = \frac{1}{(2\pi)^{r/2}} \exp\left(-\frac{u^2}{2}\right),$$

the Epanechnikov kernel is

$$K_E(u) \propto \max\{1 - u^2, 0\},$$

and the exponential kernel is

$$K_X(u) \propto \exp(-u).$$

The Gaussian and exponential kernels have infinite support, whereas the Epanechnikov kernel assigns zero weight beyond the bandwidth. The exponential kernel also decays more slowly with distance than the Gaussian kernel. Since each estimated density is subsequently divided by its mean value over the observed diffusion coordinates, multiplicative kernel normalising constants do not affect the resulting normalised density field.

Before using alternative kernels, the Gaussian KDE provides a useful consistency check on the baseline graph-degree estimator. Substituting the Gaussian kernel gives the distance-dependent term

$$\exp\left(-\frac{\|x-\psi_j\|^2}{2b^2}\right).$$

This agrees exactly with the graph-degree kernel when

$$2b^2 = h_{\mathrm{dens}},$$

or equivalently

$$b = \sqrt{\frac{h_{\mathrm{dens}}}{2}}.$$

The remaining factors in the Gaussian KDE are constant with respect to x and disappear under the mean normalisation. The normalised Gaussian KDE should therefore reproduce the graph-degree density when this bandwidth transformation is used. This equivalence is used as an implementation consistency check rather than treating the Gaussian KDE as an additional robustness specification.

The robustness analysis consequently compares the baseline graph-degree estimator with Epanechnikov and exponential KDEs. For both alternative kernels, the bandwidth is fixed at

$$b = \sqrt{\frac{h_{\mathrm{dens}}}{2}},$$

rather than being re-selected separately. This keeps the smoothing parameter fixed and allows the experiment to focus on the effect of changing the kernel form. The different kernels nevertheless have different support and decay properties, so this choice does not imply that they induce identical effective smoothing.

For each density method m, the density is normalised using its own full-sample mean,

$$\rho^{(m)}(x) = \frac{\widehat{\rho}^{(m)}(x)}{N^{-1}\sum_{i=1}^{N}\widehat{\rho}^{(m)}(\psi_i)},$$

and the corresponding potential is constructed as

$$V^{(m)}(x) = -\log\left(\max\{\rho^{(m)}(x),\delta\}\right),$$

where δ is the same density floor introduced in Section 4.4. A new density-aware graph path is then obtained for each estimator by substituting $V^{(m)}$ into the density-weighted edge cost defined in Section 4.5. The ordinary graph path is unaffected by the density estimator and therefore remains fixed.

The geometry of the linear benchmark is likewise independent of the density estimator. To ensure that changes in its minimum evaluated density cannot arise from changes in discretisation, its number of interpolation points is fixed separately for each event using the baseline graph-degree specification,

$$n_{\mathrm{lin}} = \left\lfloor \frac{n_{\mathrm{ordinary}} + n_{\mathrm{aware,baseline}}}{2} \right\rfloor$$

where n_{ordinary} and $n_{\mathrm{aware,baseline}}$ are the numbers of vertices in the ordinary and baseline density-aware graph paths, respectively. The same linear grid is then used for all density estimators.

For each method, all three path types (linear, ordinary graph and density-aware graph) are evaluated under the density field associated with that method. At observed graph vertices, density is evaluated using the leave-one-out procedure from Section 4.6. In particular, for an observed node ψ_i, its own kernel contribution is removed while retaining the full-sample normalisation scale. For a KDE this corresponds to

$$\widehat{\rho}_{-i}^{(m)}(\psi_i) = \frac{1}{Nb^r}\sum_{j\neq i} K_m\left(\frac{\|\psi_i-\psi_j\|}{b}\right).$$

Off-sample interior points of the linear path are evaluated against the full historical sample. The observed start and stress endpoints are excluded from the interior-density statistics.

The principal diagnostic is the minimum interior density,

$$M_{\min}^{(m)}(\gamma) = \min_{x \in \gamma^{\circ}} \rho_{\mathrm{eval}}^{(m)}(x),$$

which is calculated for each event and path type under each density specification. Comparisons are made within a given estimator rather than between the absolute density values produced by different kernels. The relevant robustness question is whether the density-aware path continues to attain greater minimum historical support than the linear and ordinary graph paths when the definition of density is changed.

Finally, the density-aware graph paths generated by the different estimators are compared visually in diffusion space. This distinguishes robustness of the density ordering from robustness of the path geometry itself: two estimators may support the same qualitative conclusion while assigning different density values or selecting different sequences of historical states. The resulting density comparisons and path geometries are presented in Section 5.5.

5 Macroeconomic Manifold and Stress Paths

5.1 Empirical embedding results

With the diffusion-map normalisation fixed at $\alpha = 1$ and the kernel bandwidth at $\varepsilon = 3$, the remaining choice is the number of non-trivial diffusion coordinates to retain. This determines the dimension r of the representation used throughout the subsequent analysis.

Embedding dimension. A common heuristic for selecting the embedding dimension is to inspect the eigenspectrum for a gap between consecutive eigenvalues [9]. Figure 4 shows that the most prominent gap in the present data would suggest retaining only the first two non-trivial coordinates. However, the spectral gap alone does not determine whether higher-order coordinates contain information that is important for distinguishing macroeconomic states or reconstructing the original variables.

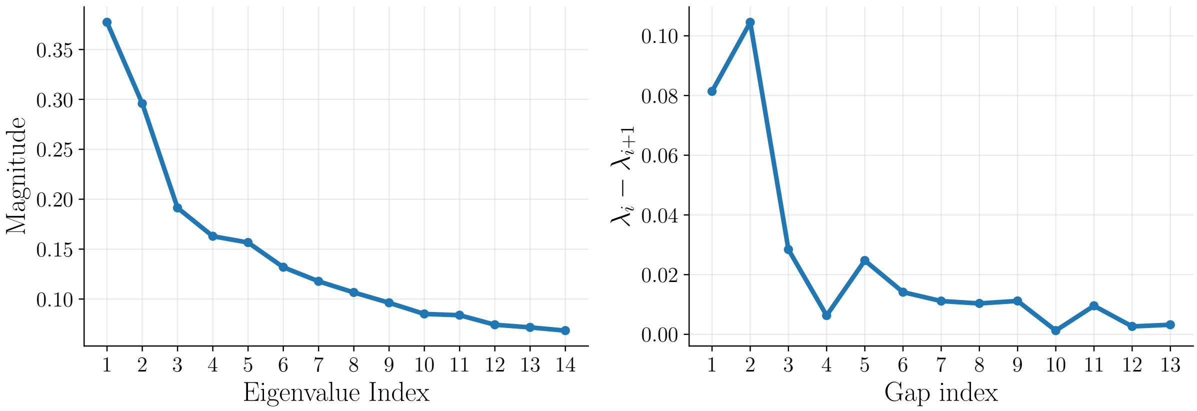


Figure 4: Leading eigenvalues of the diffusion transition matrix for the selected bandwidth $\varepsilon = 3$. The most prominent spectral gap occurs after the second non-trivial mode.

The reconstruction diagnostic introduced in Section 4.3 (make sure this is well introduced) provides evidence that additional coordinates remain informative. The proportion of variance recovered by the local-neighbourhood lifting procedure increases from approximately 30% for $r = 2$ to 45% for $r = 4$, while adding a fifth coordinate produces an improvement of only approximately two percentage points. Here, variance recovered refers specifically to the ability of the retained diffusion coordinates to reconstruct the transformed input variables, rather than to an explained-variance measure of the type used in PCA.

We additionally examine whether higher-order coordinates separate observations that appear almost identical in a lower-dimensional embedding. For each candidate dimension, we identify the closest 1% of observation pairs, excluding pairs separated by fewer than twelve months, and then recompute their distance ranks after adding one further diffusion coordinate. Table 1 reports the proportion of these near-collision pairs that no longer lie within the closest 1%, 5% or 10% of pairs in the higher-dimensional representation.

Table 1: Separation of near-collision pairs after adding an additional diffusion coordinate. Near-collisions are defined as the closest 1% of eligible observation pairs in the lower-dimensional embedding after imposing a twelve-month temporal exclusion window.

Comparison	Near-collision pairs	% Exit 1%	% Exit 5%	% Exit 10%
2D → 3D	2669	62.27	36.87	24.39
3D → 4D	2669	52.23	25.63	14.28
4D → 5D	2669	20.87	1.61	0.26

The third and fourth diffusion coordinates both provide substantial additional separation. For example, after adding ψ_3, approximately 24% of pairs that belonged to the closest 1% in two dimensions are no longer among even the closest 10% in three dimensions. A similar, although somewhat smaller, effect is observed when ψ_4 is added. In contrast, the fifth coordinate provides comparatively little additional separation, with only 0.26% of the four-dimensional near-collisions leaving the closest 10% after ψ_5 is included.

Taken together, the eigenspectrum, reconstruction performance and near-collision diagnostic suggest that the first four diffusion coordinates retain useful structure, while the incremental benefit of a fifth coordinate is considerably smaller. We therefore fix the embedding dimension at

$$r = 4.$$

Economic structure of the embedding. The resulting representation assigns each monthly observation $z_i \in \mathbb{R}^{38}$ the diffusion coordinate vector

$$\Psi(i) = (\psi_1(i), \psi_2(i), \psi_3(i), \psi_4(i)) \in \mathbb{R}^4.$$

Unlike principal component coordinates, which are linear combinations of the original variables, diffusion coordinates do not admit a direct interpretation in terms of variable loadings. We therefore investigate their statistical associations with the original macroeconomic and industry variables.

Since these relationships need not be linear, we use distance correlation rather than Pearson correlation. Distance correlation detects more general forms of dependence between variables, making it suitable for examining the nonlinear coordinates produced by the diffusion map. Figure 5 reports the distance correlation between each retained diffusion coordinate and each of the 38 input variables.

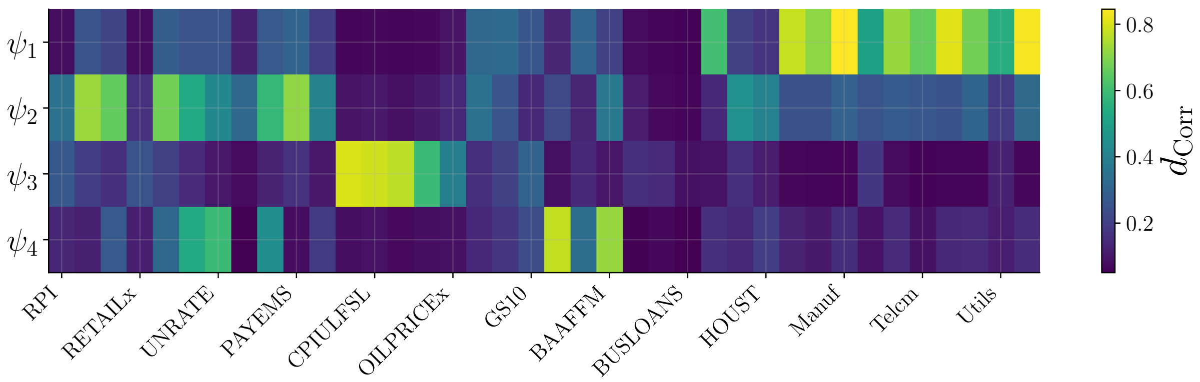


Figure 5: Distance correlations between the four retained diffusion coordinates and the 38 quantile-normalised macroeconomic and industry-return variables.

The strongest associations differ substantially across the four coordinates. Table 2 summarises the three variables with the largest distance correlation for each coordinate. These descriptions are intended only as qualitative summaries of the dominant observed associations. A diffusion coordinate is not identified with a single economic factor and may encode nonlinear variation involving several variables simultaneously.

The first coordinate is most strongly associated with the industry portfolio returns: its seven largest distance correlations are all with industry portfolios, including Manufacturing, Shops, Non-Durables, and High Technology. The second coordinate is instead associated primarily with measures of real economic activity, including real personal income excluding transfers (`W875RX1`), payroll employment (`PAYEMS`) and industrial production (`INDPRO`). The third coordinate has a particularly clear price interpretation, with its strongest associations arising from the Consumer Price Index (CPI) and Personal Consumption Expenditures (PCE) variables. Finally, the fourth coordinate is associated most strongly with the ten-year Treasury–Federal

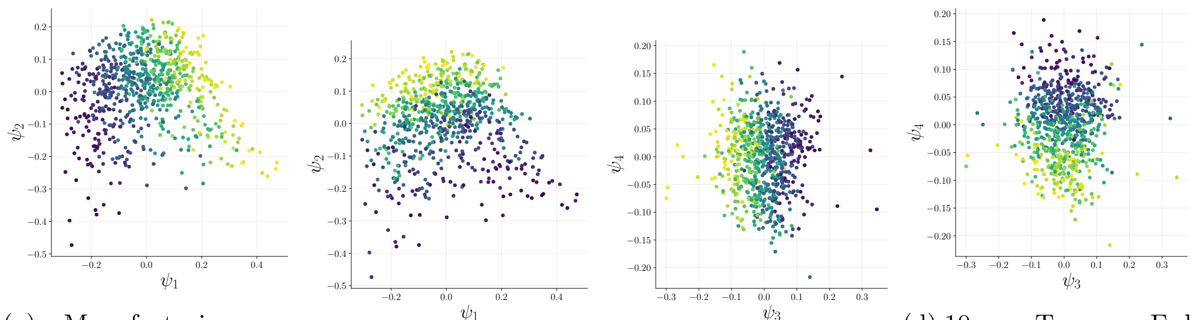
Table 2: Strongest economic associations of the retained diffusion coordinates. The interpretations are qualitative summaries and should not be read as direct factor labels. Full definitions of the input variables are provided in Appendix B. Unsure about this table, looks messy and I might be able to explain this with words well enough

Coordinate	Strongest associations (d_{Cor})	Indicative interpretation
ψ_1	Manuf (0.85), Other (0.84), Shops (0.81)	Industry portfolio returns
ψ_2	W875RX1 (0.72), PAYEMS (0.71), INDPRO (0.67)	Real economic activity
ψ_3	CPIAUCSL (0.80), CPIULFSL (0.79), PCEPI (0.77)	Prices and inflation
ψ_4	T10YFFM (0.77), BAAFFM (0.72), UNRATE (0.59)	Rates, credit and labour-market conditions

funds spread (`T10YFFM`), the BAA corporate bond–Federal funds spread (`BAAFFM`) and unemployment (`UNRATE`), suggesting that it contains information related to interest-rate, credit and labour-market conditions.

These results are also useful for interpreting the choice $r = 4$. In particular, the third and fourth coordinates contain economically distinct price and credit-rate information that would be absent from a two-dimensional representation, consistent with the reconstruction and near-collision diagnostics above.

Figure 6 provides a visual illustration of these associations. Selected two-dimensional projections are coloured by variables strongly associated with the corresponding diffusion coordinates. The colour gradients vary systematically across the embedding, indicating that the coordinates organise observations according to economically interpretable changes in the joint macroeconomic state. These plots are illustrative only; the value of a diffusion coordinate should not be interpreted as representing the value of any single macroeconomic variable.



(a) Manufacturing returns (b) Real personal income (c) Consumer prices (d) 10-year Treasury–Fed funds spread

Figure 6: Selected projections of the four-dimensional diffusion-map embedding, coloured by variables strongly associated with the corresponding diffusion coordinates. The systematic colour gradients illustrate how different aspects of the macro-financial state vary across the learned representation. I will fix this formatting!

Overall, the retained four-dimensional embedding captures several distinct and economically interpretable aspects of the joint macroeconomic and financial state. The diffusion coordinates are not treated as individually identified economic factors; rather, they provide a low-dimensional geometric representation on which the graph, density and stress-path analysis is subsequently constructed.

5.2 Embedding density results

The density estimate introduced in Section 4.4 measures the local concentration of historical observations in the diffusion representation. A point with high density ρ_i lies close to many other observed states, whereas a point with low density occupies a comparatively sparsely sampled region. We therefore interpret density as a measure of *historical support* in the learned state space.

To investigate whether this geometric notion of support is associated with economically unusual periods, Figure 7 plots the density potential

$$V_i = -\log \rho_i$$

through time together with NBER recession periods. We display V rather than ρ because this is the quantity used directly in the density-aware path cost, and because large values of V correspond naturally to increasingly sparse regions of the embedding. Since $-\log(\cdot)$ is monotonic, this transformation does not alter the ordering of observations by density.

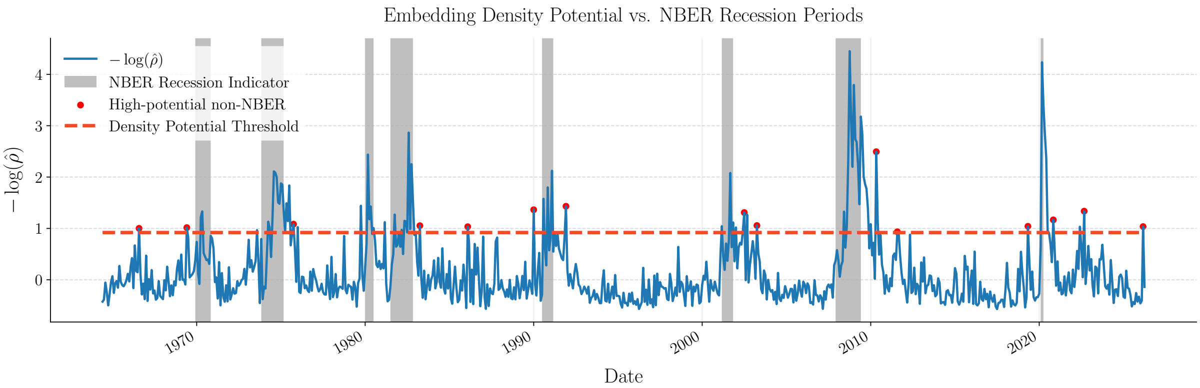


Figure 7: Density potential $V = -\log \rho$ of the observed diffusion coordinates through time, with NBER recession periods shaded grey. Larger values indicate observations lying in more sparsely supported regions of the learned representation. Red markers identify high-potential observations outside recession periods according to the reference threshold described in the text.

Figure 7 shows that elevated density potential frequently coincides with NBER recession periods. In particular, each recession contains observations for which V rises locally, indicating that several recognised periods of macroeconomic stress also occupy relatively sparse regions of the learned representation. The correspondence is not exact, however, and a number of high-potential observations occur outside NBER recessions.

To highlight these observations, we use the median density potential among recession observations as an exploratory reference threshold and identify non-recession observations whose potential exceeds this value. A four-month buffer around recession boundaries is excluded so that observations immediately surrounding a recession are not treated as separate episodes. This threshold is used only as a descriptive device rather than as a formal classification rule. The presence of high-potential observations outside NBER recessions is not unexpected: the NBER regions provide a binary classification of business-cycle recessions, whereas the density measure identifies unusual configurations of the complete joint macroeconomic and financial state. Low historical support in the diffusion representation should therefore not be interpreted as synonymous with recession status. I think the last point is important, but I might get rid of the highlighted points. I intended to investigate each of them and show how they corresponded to some sort of economic outlier, but I ended up just fishing for explanations.

As an additional check, we compare diffusion-space density with local support in the original 38-dimensional quantile-normalised input space. For each observation, we calculate its average distance to its 15 nearest historical neighbours in this space. The distance correlation between diffusion density and this nearest-neighbour distance is 0.45, while the Pearson correlation is -0.46. The negative Pearson correlation indicates that observations assigned high density in diffusion space tend, on average, to have closer neighbours in the original transformed data. The magnitude of the relationship is moderate rather than perfect, which is consistent with the diffusion map retaining a low-dimensional representation of the data.

Together, these results provide evidence that the density estimate has a meaningful interpretation beyond the diffusion coordinates themselves. High-density regions correspond to states with greater local historical support, while unusually sparse regions frequently coincide with recognised periods of macroeconomic stress. This motivates the use of the density potential in the stress-path construction considered in the following section.

5.3 Stress-path results

The events and dates in Table 3 were selected based on both their macroeconomic variables and news articles indicating important historical dates. For our benign states, we inspected several macroeconomic variables to ensure that the economy was not in a strange state. Obviously not sure about this. Do I need to show anything for my choices? I looked at a handful of macro variables at each state. I also researched when the crises actually occurred. For the stress points, we compared the macroeconomic variables and various news sources to identify regions of genuine economic stress. For the investigated five pairs, however, only four were retained due to one pair being too close in diffusion space. This choice was made before further testing commenced to avoid bias. The chosen endpoints are displayed in Table 3, and an in-depth discussion of the dates is included in Appendix A This is actually not in the appendix. Should I include it?.

Table 3: Macroeconomic Regimes and Event Analysis

Regime / Macro Event	**Start Date**	**End Date**
Pre-Crisis to Early 1980s Recession Trough	1977-01-01	1982-07-01
Benign Expansion to Global Financial Crisis (GFC)	2006-03-01	2008-10-01
Benign Expansion to COVID-19 Shock	2019-07-01	2020-04-01
Low-Inflation to Monetary Tightening	2019-04-01	2022-04-01

For each of the events in Table 3, we identify the indices corresponding to the pair of dates. Using these dates we can find which nodes correspond to each stress state pair in diffusion space. Then we calculate our three types of paths. First we linearly interpolate between the two points through the diffusion space. Next, we find the ordinary graph path through the kNN graph build from the diffusion coordinates. Finally, we use our density-aware graph cost to find a density-aware graph path through diffusion points. In Figure 8, we plot the resulting paths for each scenario on a 2D projection of our embedding.

In the top left of Figure 8, we can clearly see the density-aware path deviating towards the high-density region of diffusion space. However, if we look at both of the figures on the right, it appears as if the density-aware paths are actually doing the opposite. This is because the two-dimensional projection does not capture the full structure of the embedding. To genuinely compare the paths, we rely on the metrics described in Section 4.6. Firstly, instead of trying to understand the density of the regions a path passes through by looking at the two-dimensional projections, we can plot the density of each point along a path (Figure 9).

We can make a variety of observations by looking at Figure 9. Firstly, the density-aware path always attains the maximum density, and also has the highest *minimum* density across all paths. Each event actually tells a different story, not sure if I should go into this much detail for the 1982 recession event, we see that the density-aware path vastly increases the density along the path compared to the ordinary graph path and the linear path, whose density profiles looks very similar. For the 2008 GFC event, the graph paths share similar density profiles while the linear paths profile is far lower at all but one interior point. In the 2020 COVID event, there is a clear hierarchy, where the linear path has the lowest density profile, followed by the graph path, followed by the density-aware graph path. Finally, for the 2022 monetary tightening event, all of the paths have a very similar density profile.

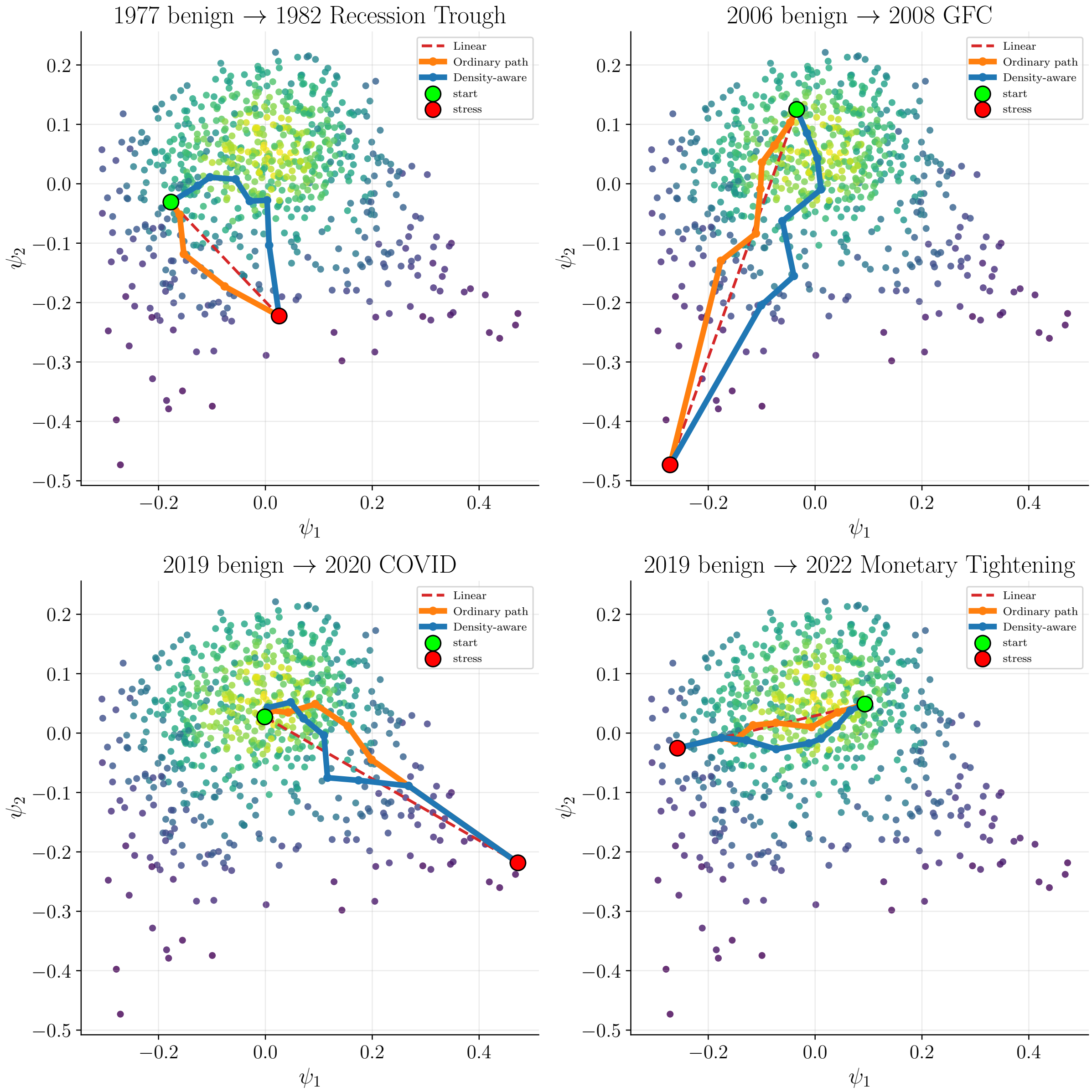


Figure 8: Put one legend on the bottom or one for each plot?

While it might be tempting to try to explain these results by looking at Figure 8, this is not appropriate as the graph paths plotted in different coordinates can look extremely different. Should I include a figure of this in the appendix? Or even here? We can, however, note that in the events where the stress events have an extremely low density (GFC and COVID), the linear paths have a much lower density profile than the graph paths. On the other hand, for stress points which have a reasonably high density proxy (should I define 'high'? median value? percentile?), the linear path seems to have a density profile at least similar to the graph path. This is an important finding as it suggests that ordinary graph paths may only improve the density score, and thus historical support (how interchangeable are these? and which one should I say?) when compared to a linear path when the stressed endpoint is an extremely sparse region of diffusion space. Otherwise, a linear path through diffusion space manages to remain within a high-density region without cutting through regions of low historical support.

Now we want to compare the differences in the 1982 Recession and the 2022 monetary tightening event. Why does one show the density-aware path performing much better compared to the ordinary graph path, while the other seems to perform similarly to even the linear path? We refer to Table 4 that the 1982 recession path starts in a relatively low-density region and ends in a slightly lower-density region, whereas the 2022 monetary tightening path starts in

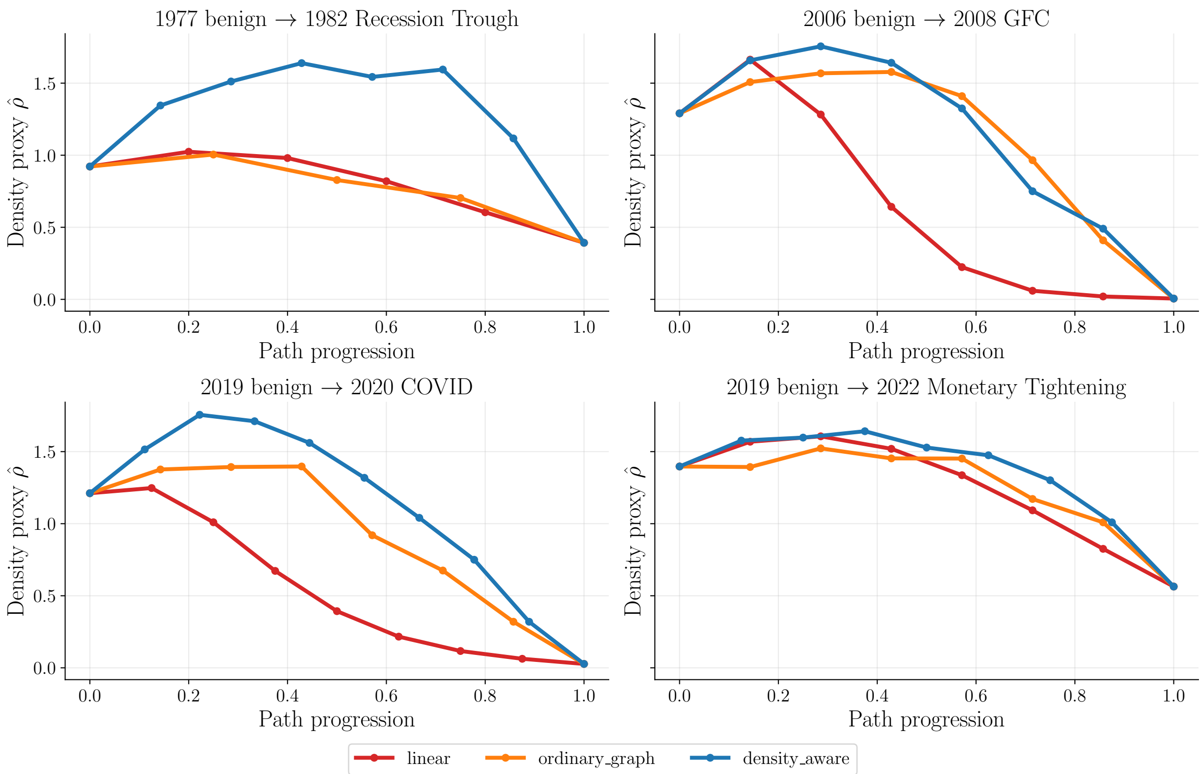


Figure 9: Caption

Table 4: Endpoint densities for the 1980s recession and the 2022 monetary tightening events. DO I INCLUDE ALL ENDPOINTS?

Regime / Macro Event	Benign Density Percentile	Stress Density Potential
1980s Recession	39.43	11.17
2022 Monetary Tightening	81.16	19.11

a very high-density region and ends up in quite a high-density region for a stress point. Not sure how much of the next bit I can really say. I am saying what i think, but it is guided by guessing/using what I see in the graphs, which I said we shouldn't do This might suggest that in the 2022 monetary tightening event, we are cutting through the middle of a high-density region to just outside of the high-density core (I've not really defined the 'core', how do you know that there is one?), meaning that the ordinary graph path and density-aware path both take a similar path, as the density penalty does not negatively impact the cost here. On the other hand, for the 1982 recession trough, we start in a relatively low-density region and end up in another low-density region, so the density-aware path may be able to make a big detour towards the high-density core in order to minimise the density-aware objective function. This is what we see in Figure 8, however, we should not base our conclusions on this alone, due to the impact of the different projections. We can instead investigate how the density penalty impacts the path in diffusion space by measuring the total path length and the total turning angle of the path. We present those results in Table 5.

In Table 5 we can see that the total turning angle for the density-aware path in the 1982 recession event is significantly higher than the ordinary graph path for the same event, this implies that the density-aware path is siginificanlty changing direction in order to minimise the density-aware cost. The L_{diff} metric also backs this up showing that the density-aware path is 1.4 times longer than the ordinary graph path. Here we see the modified cost function in action, where the density-aware graph path is actively taking a longer route to avoid regions of sparse historical support.

Table 5: Stress-path evaluation across the four historical scenarios. Higher values of $M_{\min}$ and M_{mean} indicate greater historical support. L_{diff} is measured in diffusion-coordinate space and D_{turn} is the total turning angle.

Scenario	Path	$M_{\min}$	M_{mean}	L_{diff}	D_{turn}
	Linear	0.60	0.82	0.29	0.00
1977–1982 recession	Ordinary graph path	0.70	0.74	0.35	2.45
	Density-aware	**1.12**	**1.24**	0.48	5.51
	Linear	0.02	0.65	0.78	0.00
2006–2008 GFC	Ordinary graph path	0.41	0.71	0.84	4.03
	Density-aware	**0.49**	**0.78**	0.88	5.23
	Linear	0.06	0.54	0.64	0.00
2019–2020 COVID-19	Ordinary graph path	0.32	0.66	0.69	3.72
	Density-aware	0.32	**0.85**	0.85	6.73
	Linear	0.83	1.28	0.37	0.00
2019–2022 tightening	Ordinary graph path	1.01	1.26	0.45	5.96
	Density-aware	1.01	**1.33**	0.46	4.96

For the 2022 monetary tightening event, the density-aware graph path has a very similar length to the ordinary graph path and the total turning angle is actually less than that of the ordinary graph path. This suggests that the region traveled by these paths is not impacted by the density-aware cost.

In both of these events, the mean density of the linear path is actually higher than that of the ordinary graph path. Suggesting that simply using ordinary graph paths through diffusion coordinates does not necessarily avoid weakly supported regions.

We also see that for the 2020 COVID event and the 2022 monetary tightening event, the minimum density of the ordinary graph path and the density-aware graph path is the same. This simply reflects the paths choosing the same node at a certain point (likely the second last node in the path).

We see that the weighted average of the graph path density is always the highest in each event, suggesting that this method is effective at improving historical support for stress trajectories.

Would it be good to talk about the actual cost function again here to remind people / show people who've skipped to the results?

$$w_{ij}^{(\beta)} = \|\psi_i - \psi_j\|^2 \exp\left(\beta \frac{V_i + V_j}{2}\right).$$

5.4 Lifted-path results

We compared two ways to lift linear paths into macro-financial space. The first method described in [3] uses a simple linear operator H, the second called a local-neighbourhood lift used the weighted average position of the nearest 30 neighbours to a point in diffusion space to map it to the macro-financial space. We compared both methods and found their performance to be very similar (performance at what?). Going forward, we use the local-neighbourhood lift as it performed marginally better when comparing the distances from trajectories between stress points to other observed points. The full analysis is in Appendix D.

The aim of this dissertation is to generate paths along the learned geometry that remain historically supported, not directly temporally reasonable. This is worth baring in mind when looking at the following figures, where we plot a selection of macroeconomic variables along generated stress paths. We also plot the real path of the macroeconomic variable to compare. While we don't directly expect the generated path to follow the real trajectory, if one does then we can interpret that as the path following a historically supported path.

In this dissertation, we have focused on generating paths in diffusion space and have made adjustments there to try to enforce that our paths remain in historically supported regions (Is there a way we can measure that in macro space? Maybe discuss the fact that we haven't done this.) In this section we briefly show what our density-aware paths look like in terms of a selected few macroeconomic variables. We do this so we can see what our graph paths make any sense in the real world. We also plot the lifted linear path as a comparison, and the observed path taken by the real economy between the endpoints.

The following paragraph doesn't really make sense and is very hypothetical. I need to say something along these lines. Path jump != time, this is a limitaiton. Also means that comparing to the observed path doesn't tell us much Travelling through the diffusion coordinates is not intended to give us paths which reflect any sort of timescale, so comparing against the observed paths needs to be carefully considered. A graph path should represent a way to go from one state to another in such a way that all of the combinations of macroeconomic variables are possible and that we specifically tune towards highly supported combinations. This must be taken into consideration (Should it?) because, for example, the real path in the COVID event has 9 observations, which is the same as the number of observations in the density-aware graph path; however, the number of observations in the 2022 monetary tightening event is 37, whereas the density-aware and linear path only has 9 points. So while the graph paths haven't been explicitly given an awareness of time, their affinities implicitly contain some temporal information because observations of close-by months may be close together, therefore giving high affinity. So then neighbours close in diffusion space may be close in time. This is important because in the COVID event example, we have asked the model to find us a path from two endpoints while minimising the distance of the path. This means that finding a long path going through many nodes may not actually make sense over the long time scale. The graph path is likely has some sort of idea that a jump in the diffusion coordinates can be completed in roughly a month. Then, if we want to find a path from two points which happens over many months, the graph only has 10 jumps; then comparing the two paths may not make the most sense. This could be remedied by finding a notion of time between jumps and then imposing the path to have a certain number of jumps depending on how many months you want the path to represent.

Event	Months between endpoints
Recession	66
GFC	31
COVID	9
Tightening	36

Table 6: Caption

I really don't know how much analysis of this we need to do Saying this, we can see that in Figure 10a our path does not fluctuate as much as the FEDFUNDS (Federal Reserve interest rate) macrovariable does. The observed path is over 66 months, the biggest gap of all the chosen events. It therefore makes sense that the graph path doesn't fluctuate that much, because that would mean that the graph path would be visiting low-density nodes between the endpoints. The linear path also stays relatively stable throughout this path I want to say because it goes directly to the one stress point, without going through different regions of diffusion space which correspond to the other stress poitns along the way. I don't know if I can say this. It is also worth noting that the interest rates during this period were the highest recorded in the FRED database [13] and so were the corresponding monthly changes in the rate. For the UNRATE (unemployment rate) we see the linear path stay very constant at a high unemployment rate, not sure what else to say about this.

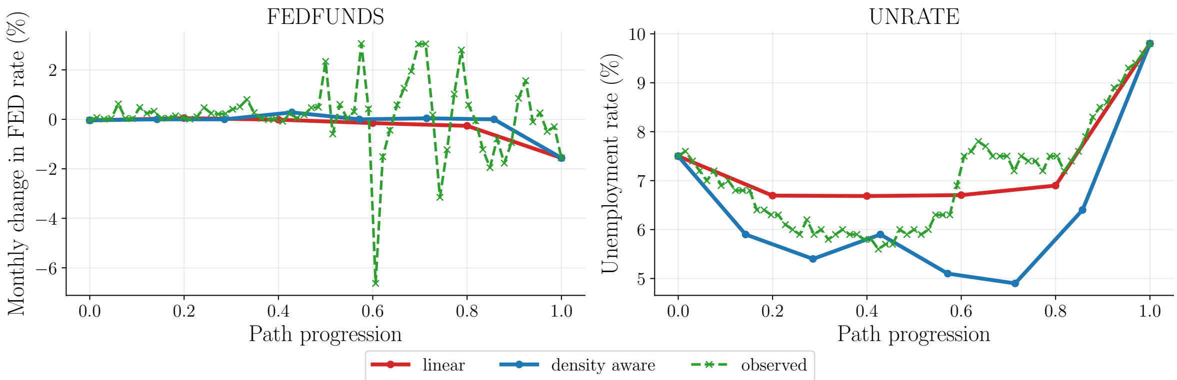
In Figure 10b the linear path and the density-aware path seem to be quite similar. Both seem to have a general trend towards the stress points (the linear path for BAAFFM (BAA rated corporate yields minus the FED rate) may deviate from the observed path slightly more - I don't know if I should be saying that the closer to the observed path the better), but we know from the density of each path that for each point in the density-aware path, the rest of the macroeconomic and financial variables are in a combination which is much more historically supported. So while these two paths look similar (I am kind of looking at 4 paths, linear+density-aware for HOUST and BAAFFM) we know that the combinations of the rest of the variables are historically supported for the density-aware path and not for the linear path. I think this is important to remind people of, the paths don't necessarily look good - and we don't necessarily want them to look identical - but the combination of the rest of the variables is supported.

In Figure 11a, we see an example of a graph path whose number of steps is equal to the number of steps in the observed path. (This still doesn't mean they are directly comparable, I think, but it might let us get a better view??) We can see in both cases that the linear path gradually increases the values of the macroeconomic variables, whereas the graph path remains flatter until the shock where the path then jumps up to the stressed value. This is more of a similar manner to what the observed path does. (I am not sure if this is worth talking more about?).

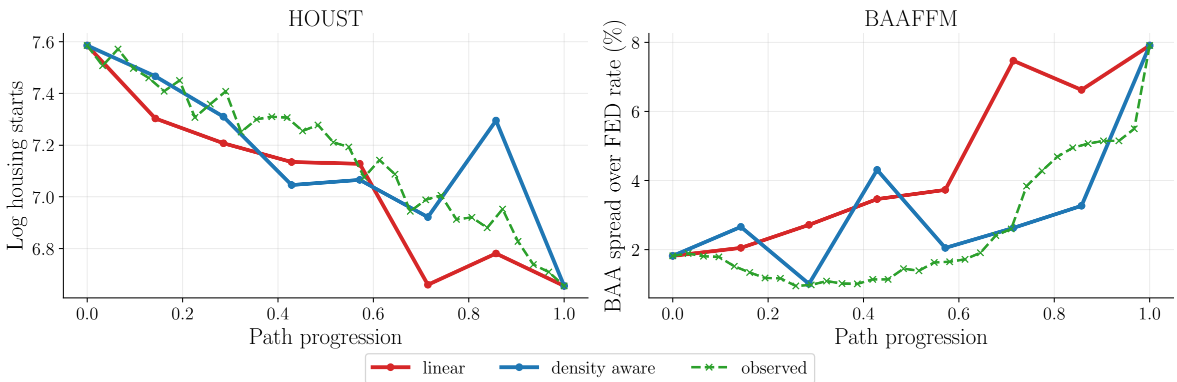
In Figure 11b we can see that the density-aware paths are more jagged than the linear path which is very straight. This represents the density-aware path ensuring that each point along the path is an observation, which means that it cannot take the straight path. It may represent the combinations of variables changing in such a way that they remain historically supported (It does represent this doesn't it?). The spike we see in the FEDFUNDS plot here may actually be insightful as we see the observed path also had a spike (albeit in the opposite direction), but this could represent the way the variables need to organise themselves in order to get between endpoints.

I don't know how much I should talk about these points. I think it's important, but we can only really look at the graphs; we can't really say much else. Maybe we should link it to the density more.

I don't actually think that there should be that much discussion around this. We work in the diffusion coordinates in the hopes that we get good paths here, but we don't actually come up with any tests to show that they are good paths in macro space. I think this section may just be for visualisation.

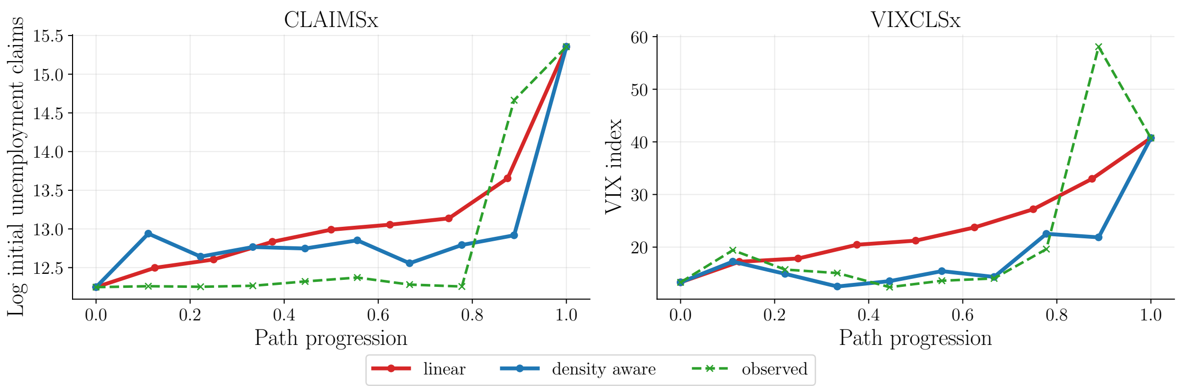


(a) Transition from the benign state in 1977 to the 1982 recession trough.

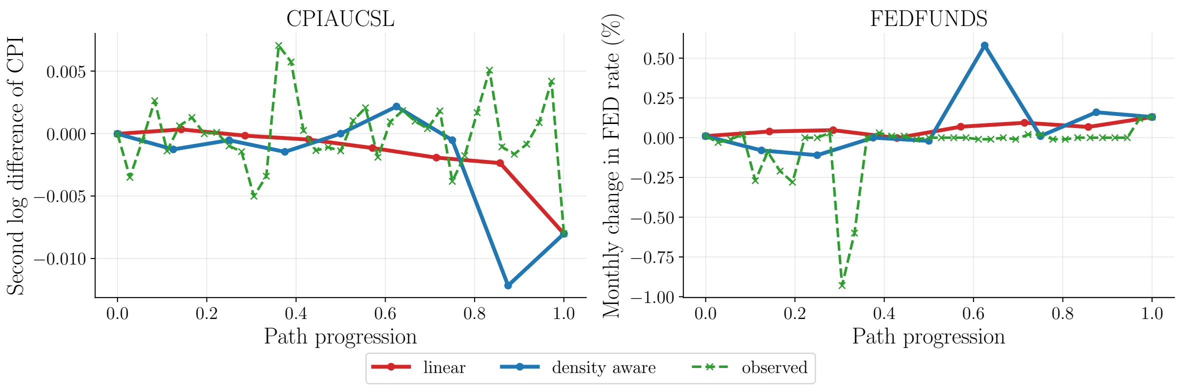


(b) Transition from the benign state in 2006 to the Global Financial Crisis in 2008.

Figure 10: Lifted macroeconomic trajectories for the early-1980s recession and the Global Financial Crisis. The graph path consists of observed transformed macroeconomic states, whereas intermediate points on the linear diffusion-coordinate path are reconstructed using local neighbourhood lifting.



(a) Transition from the benign state in 2019 to the COVID-19 disruption in 2020.



(b) Transition from the benign state in 2019 to the monetary-tightening episode in 2022.

Figure 11: Lifted macroeconomic trajectories for the COVID-19 disruption and the 2022 monetary-tightening episode. The graph path consists of observed transformed macroeconomic states, whereas intermediate points on the linear diffusion-coordinate path are reconstructed using local neighbourhood lifting.

5.5 Robustness of parameters

In this section we examine what happens to our main result (that density-aware paths avoid sparse regions more than linear paths and more than ordinary graph paths Is this the main result? Make sure this is obvious) as we change the following parameters:

$$\beta, \quad k, \quad \varepsilon$$

where β controls the strength of sparsity avoidance of the density-aware graph path, k controls the neighbourhood size of the kNN graph built from the diffusion coordinates $\Psi(i)$, and ε is the bandwidth used to create the affinity matrix

$$W_{ij} = \exp\left(-\frac{\|z_i - z_j\|^2}{\varepsilon}\right).$$

why only these parameters? why not the rest? Also, justify why you don't look at the other metrics like min density and turning angle/length

Firstly, we vary the value of β and monitor the weighted mean density of the density-aware path. This is the only path that we need to monitor, because varying β does not impact the density of the ordinary graph path or the linear path.

In Figure 12, we can see clear stability as we vary β. We can also see that setting $\beta = 0$ coincides with the ordinary graph path. Increasing β tends to increase the weighted mean path density, although there are many intervals where a change in β does not change this metric. We also see that for the 2008 GFC event, increasing β from 5 to 6 actually decreases the weighted mean density of the density-aware path. This sounds counterintuitive at first; however, the density-aware graph path is not constructed to maximise the weighted mean density; it is constructed to minimise the objective

$$w_{ij}^{\beta} = \|\psi_i - \psi_j\| \exp\left(\beta \frac{V_i + V_j}{2}\right)$$

for all intervals in the graph path. Do I try and come up with an example of how this would happen or just say that it is possible because the density-aware shortest path doesn't seek to maximise the mean density. But we have been using the β parameter to try to increase the mean up to now, so it would seem a bit of a cheat to say "well, it was never specifically designed to do that, so this is not shocking"

Next, we varied the k parameter in the kNN graph construction from the diffusion coordiantes. Do I have to say why I do this?. Figure 13 shows a similar story to Figure 12, where our result is consistent: density-aware paths have higher historical support than ordinary paths and linear paths. There is one exception to this, however: the 2020 COVID event. In this event, for $k = 5$, the weighted mean density of the ordinary path is slightly higher than that of the density-aware path. I don't know why this is. Am I supposed to speculate or actually find the reason? Or just mention it, saying that this value of k might not be suitable for some reason?

Now we investigate the impact of changing the bandwidth parameter ε when building the diffusion map. This will impact every downstream result, including the diffusion density estimate. The density estimate is normalised in each case, which makes them of a similar scale; however, they are not directly comparable. We therefore present the results in Figure 14 as a panel of bar charts. In this figure we can see that our results are generally stable across all nearby bandwidth parameters, suggesting that this method is robust. (Do I need to say something like that? I feel like showing all of these results and then moving on is not worth it). In one event (2022 monetary tightening), we find that the linear path actually has a higher density than both the ordinary graph path and the density-aware graph path. As discussed in Section 5.3, this event connects two endpoints directly through a very dense region (I don't know if I made this explicitly clear, nor do I know if I actually can). Due to the leave-one-out density calculation, which disregards the current graph point, there is no extra density score given to a node if there

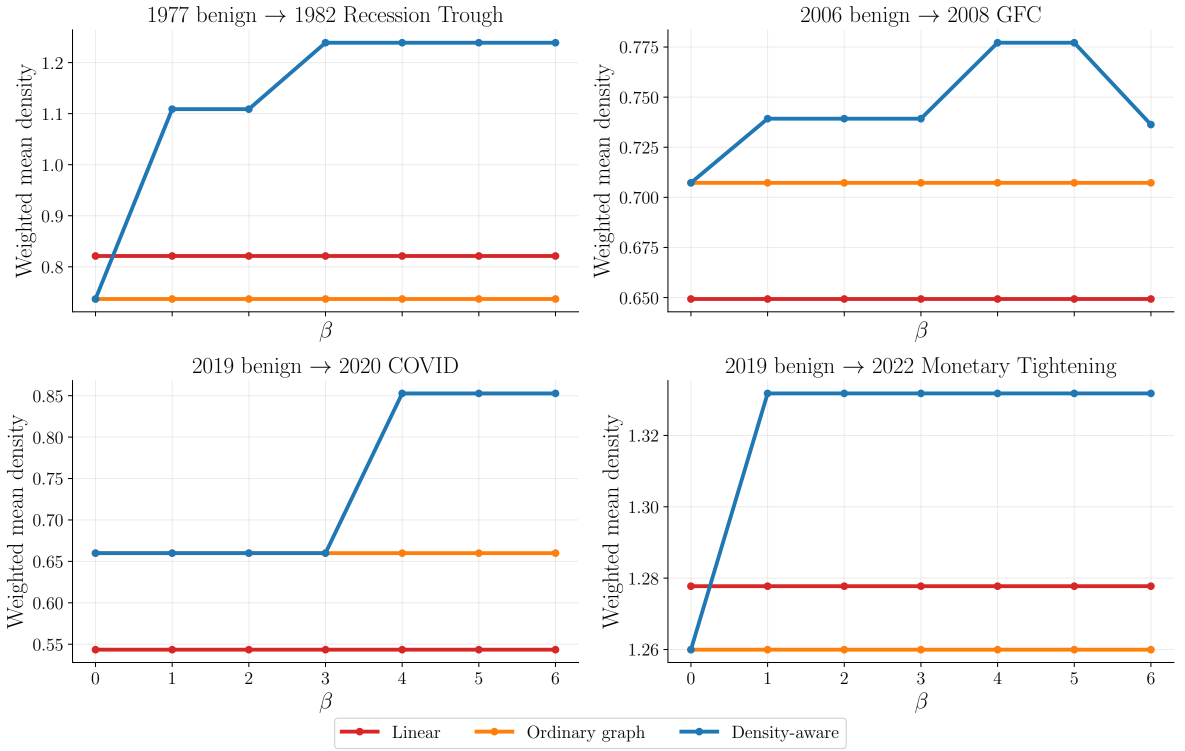


Figure 12: Caption

Figure 13: Caption

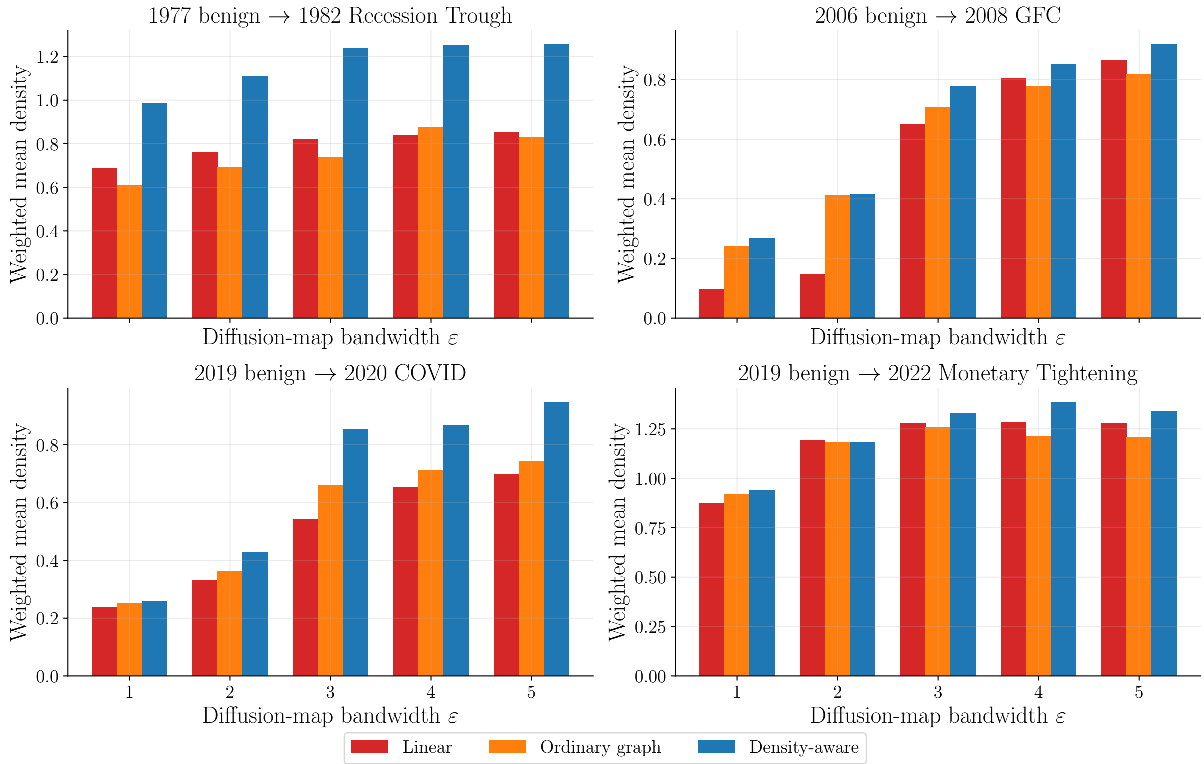


Figure 14: Caption

actually is an observation in that location. In a high-density region, where lots of points are close together, it makes sense that a linear path through this region is assigned a high density, and there isn't much to say that the graph paths are guaranteed to have a higher density in this region just because they follow observed nodes, due to the leave-one-out calculation. (I think this is a bit crap). Why does changing ε impact this? Looks like it just shifts things around, so if your previous explanation works well enough, you can just say that because the bandwidth slightly changes the distribtion in diffusion space, now the linear path is closer to observed nodes than the nodes in the graph paths. In this event, the minimum density of both graph paths is higher than the minimum density of the linear path. This is maybe important to say, but I don't know how much I should talk about the robustness of the other metrics if I don't properly introduce them.

Finally, we explore what happens when we change the density estimation function. We compare our results to using kernel density estimates with 2 different kernels. We used the exponential kernel and the Epichinikov kernel to estimate density. The exponential kernel gives more weight to farther-away points compared to the Gaussia kernel used in the graph-degree calcualtino that we used, i.e. it has heavier tails. Whereas the Epichinikov kernel has finite support and assigns zero weight to points for which $\|\psi_i - \psi_j\| > 1$. Using the new density estimates, we calculate new density-aware graph paths between endpoints. We also must re-evaluate the density of points along the linear paths and ordinary graph paths, even though the paths themselves do not change under the new density estimates. We compare the weighted mean density of the linear path, ordinary path, and density-aware path in Figure 15.

For only one event, the ordering of the path types has changed. For the exponential kernel in the 2020 COVID event, both graph paths follow the exact same nodes, leading to the weighted density being identical. In the rest of the events, for all of the kernels, the result remains the same: density-aware paths travel through regions of higher historical support than both linear paths and ordinary graph paths.

Do I need to conclude this or not?

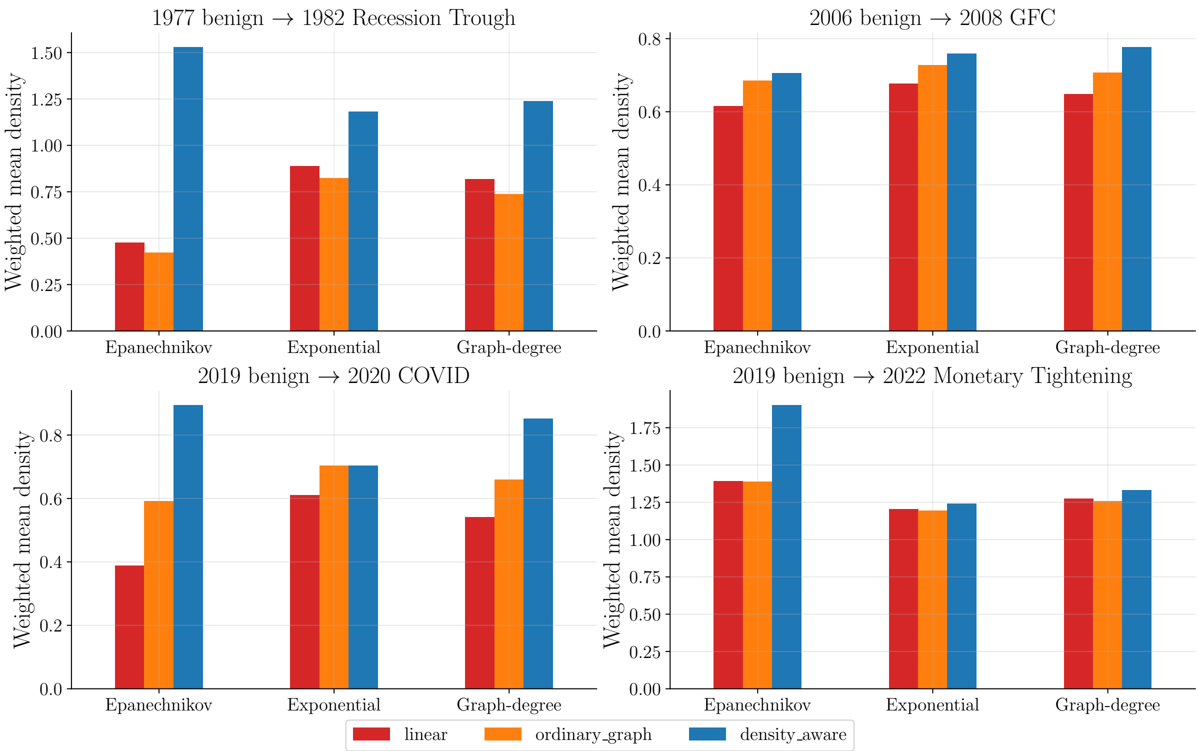


Figure 15: Caption

6 Markov bridges extension

6.1 Markov bridge motivation and background

A natural extension is being able to find a distribution of paths. In modelling, you rarely want to return a deterministic answer; rather, you would like to see a distribution of scenarios with corresponding probabilities. Therefore, the question is, can we generate a distribution of stress paths between two endpoints which follow the learned geometry? This section introduces a method which can be used to generate a series of stress trajectories which travel along the learned geometry. We also introduce a density-aware methodology which is analogous to the main part of this dissertation. The aim of this section is twofold: to implement a way to generate sample paths which travel along the learned manifold and avoid weakly historically supported regions, and to strengthen our findings of the density-aware graph paths, by seeing whether they represent a path along the whole graph random walk by using only the leading eigenvalues of the transition matrix.

In our construction of the diffusion map, we create a transition matrix P, whose elements P_{ij} correspond to the probability of node i jumping to node j in a single step of a graph random walk. We can extend and find that $(P^k)_{ij}$ is the probability that node i jumps to node j in k graph random walk steps. This probability matrix contains information about the underlying geometry because of the bandwidth parameter we chose to uncover the local connections.

A Markov bridge is constructed by conditioning a Markov process to start at a specific point and end at a specific point over a certain number of random walk steps. We can then get a series of random variables

$$X_0, X_1, \ldots, X_K,$$

where K is the number of steps in the Markov bridge, and X_0 and X_K are the conditioned start and end points, respectively.

These paths then represent possible paths through observed points from the start state to the stress state. The jumps are not bounded by the k nearest neighbours like in the density aware graph paths what are the implications of this? This means things are jumping off the manifold in our jump, like setting k too high in the graph. The long jumps have low probabilities, and could happen in real life potentially. Worth testing with the computed length of jumps for consecutive nodes in our sample?. The aim of our work was to build paths between endpoints, through highly supported regions. The Markov paths only have a notion of local connectivity, not density. This is quite correlated, I would assume; the high-density points have highly connected neighbours, e.g. Then instead of taking simple Markov bridge paths, we introduce a density-aware Markov bridge path via

$$\bar{P}_{ij} \propto P_{ij} \exp(-\alpha V_j),$$

where V_j is the density potential introduced in Section 2.3 and α is a strength of sparsity avoidance. This has the effect of reducing the probability that node i jumps to node j if node j is a low-density node.

Here you can maybe explain more background on Markov bridges We are using a different transition matrix now, same method as in the construction in the Laplacian eigenmap paper [4]

6.2 Markov bridge methodology

To sample from a Markov bridge, we don't need to generate the entire step-dependent transition matrix Q_k; instead, we can introduce

$$h_k(i) := \Pr(X_K = S | X_k = i) = (P^{K-1})_{iS},$$

which tells us the probability that we will reach the required stress state, given that we are currently at node i. These vectors allow us to condition our transitions via the following

$$\Pr(X_{k+1} = j | X_k = i, X_K = S) = P_{ij} \frac{h_k(j)}{h_{k-1}(i)}.$$

We can see that, in our Markov bridge, the probability of going from node i to node j is increased if the probability of going from node j to node S in the remaining steps is high (relative to the other nodes). To generate a path, we start at a specific node i and use this formula to figure out the probability that we transition to the next node j. We can do this until we reach S and repeat for however many simulations we want.

We build the vectors h via backwards recursion with the end condition that

$$h_K(i) = \begin{cases} 1 & \text{if } i = S \\ 0 & \text{if } i \neq S. \end{cases}$$

This condition says that the last node in the chain must be the stressed node Why does it say that? Because of what we defined above with the $P\frac{h_1}{h_0}$ thing. We work backwards from this to find h_{K-1}. We remind ourselves that

$$h_{K-1}(i) = \Pr(X_K = S | X_{K-1} = i),$$

in words: the probability that we end up at the stressed node, given that we have got one step to get there. This simply corresponds to the probability of jumping from any node to S which is $P_{:,S}$. This is equivalent to Ph_K, therefore

$$h_{K-1} = Ph_K,$$

thus the start of the recurrence relation. For illustration purposes, we explain how $h_{K-2} = Ph_{K-1}$. We want to know the probability of landing at the stressed node given that we have two steps of our random walk remaining. We know that h_{K-1} tells us the probability of finding a path from any node to the stressed node in one step. Then we can find the probability of arriving at any of these nodes in the previous step. In a specific row i of h_{K-1}, then, we have the sum of the probabilities that we start at node i and find a path through to S. This is exactly Ph_{K-1}. I don't think this is very helpful. We generate and store the vectors h_k for each step.

We also store the marginal probabilities. We use this for the entropy calculation and the probability mass calculation

To sample a path conditioned to start at node B and end at node S, we use a combination of our transition matrix and our vectors h_k. Firstly, we know that $X_0 = B$, so we can find

$$\Pr(X_1 = j | X_0 = B, X_k = s) = P_{ij} \frac{h_1(j)}{h_0(B)},$$

vectorised we have

$$Q_1(B, :) := P_{B,:} \frac{h_1(:)}{h_0(B)}.$$

Then we sample

$$X_1 \sim Q(B, :),$$

giving us the next point in the Markcov bridge. We repeat the process finding

$$Q_2(X_1, :) = P_{X_1,:} \frac{h_2(:)}{h_1(X_1)},$$

and sampling

$$X_2 \sim Q_2(X_1, :).$$

We continue sampling until we reach X_K, which, by construction, is always our stressed endpoint. Repeating this process 10,000 times gives a distribution of paths from our benign to stressed point through the learned geometry.

To create our density-aware transition matrix, we simply create a column vector whose elements are $L = \exp(-\alpha V_j)$, and we multiply every column in P by the corresponding element in L. We then normalise this matrix so that the each row sums up to one, thus defining a new transition matrix $\bar{P}$, where $\bar{P}_{ij}$ is the probability of node i jumping to node j in the density-aware scheme.

The only notion of the diffusion coordinates these Markov bridges have is the fact that their transition probabilities are varied by the density of the diffusion coordinates. However, the diffusion coordinates represent eigenfunctions of this diffusion process Do they?. We examine the structure of these graphs by mapping them to the corresponding diffusion coordinates. This is so that we can compare them to our density-aware graph paths And I think that it is quite natural because of how the diffusion coordinates are defined by this matrix anyway, so there is a strong connection.

For each event, we compute the density-aware path, and then we compute the Markov bridges, where we set the number of jumps in each Markov bridge to be the same as the number of jumps in the density-aware path for that event. This allows us to directly compare each of the computed metrics.

We compute the same metrics as we did in Section 4.6, namely we calculate

$$M_{\text{min}}(\gamma), \quad M_{\text{mean}}(\gamma), \quad L_{\text{diff}}(\gamma), \quad T_{\text{turn}}(\gamma),$$

for each generated path. This results in a range of values. From this range, we inspect the distributions and compare them to our density-aware graph path.

We also compute two distribution-specific metrics, which we cannot compute for the deterministic density-aware path. Firstly, we calculate the entropy of the bridge paths, using the marginals $\mu_k(i)$ where k is the step in the Markov bridge and i is the node index. The entropy is defined as

$$H_k = -\sum_{i=1}^{N} f(\mu_k(i)),$$

where

$$f(\mu_k(i)) = \begin{cases} \mu_k(i) \log \mu_k(i) & \text{if } \mu_k(i) \neq 0, \\ 0 & \text{if } \mu_k(i) = 0. \end{cases}$$

Citation? The marginals $\mu_k(i)$ tell us the probability of the random variable k steps along the Markov bridge being node i. The tighter the spread of the Markov bridges, the smaller the entropy will be. Since $0 \leq \mu_k(i) \leq 1 \ \forall k, i$, then $\log \mu_k(i) \leq 0$ and $H_k \geq 0$. We compare the entropy of the Markov bridges for $\alpha = 0$, and $\alpha > 0$.

Next, we calculate the expected distance of a Markov bridge path to the density-aware path via

$$d_k = \mathbb{E}\left[\|\Psi(X_k) - \Psi(\Gamma_k^\beta)\|\right],$$

however, because we have stored the marginals, we can calculate this as

$$d_k = \sum_{i=1}^{N} \mu_k(i) \|\psi_i - \psi_{\Gamma_k^\beta}\|.$$

Finally, we can use the Markov bridges to generate distributions of paths in macroeconomic space where each point in every path corresponds to a historically supported combination of variables. We plot a confidence interval of a selection of macroeconomic variables for each stress event.

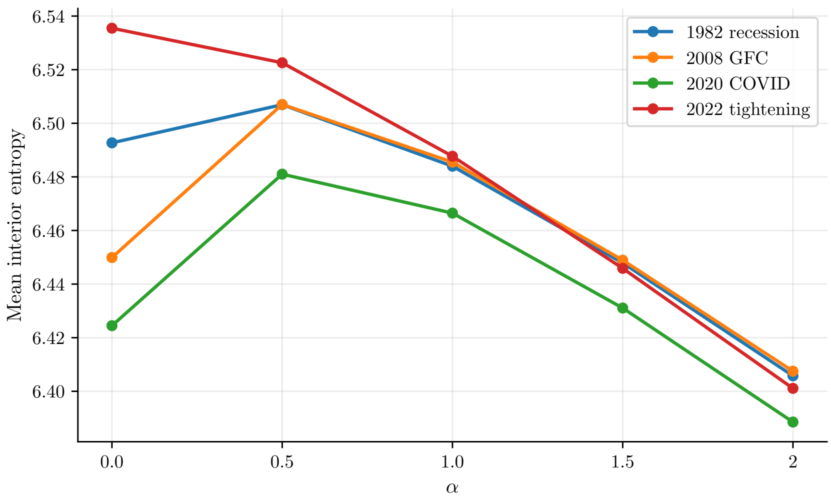


Figure 16: Caption

6.3 Markov bridges results

We first read the results of the entropy of the Markov bridges over an increasing value of α. Increasing α decreases the probability of jumps to low-density nodes, and therefore, one would expect the variation of the distribution to decrease as α increases. In Figure 16 we see exactly that. In Figure 16 we can see that increasing α has the effect of making the distribution of the Markov bridge much tighter (less variance). This is exactly what we expected and it's exactly what we desired. A slight worry is that the entropy of each of the distributions looks very similar across events, I am not sure if this is what you'd expect?. This might suggest that regardless of the endpoints, the distribution of the interior of the Markov bridge is similar, which I don't think this would be desired. Maybe this is how the economy works?. I have checked and it doesn't look too bad. The means of the paths have different points; these points get closer together and further into the central high-density region as α increases, implying that increasing α so much means that the chain quickly forgets about the endpoints, it mainly just makes jumps between high density points and then makes another big jump to the endpoint when it needs to. This could be remedied by varying the density calculation to have a wider spread of high-density points, meaning that while avoiding spare regions, the Markov bridge could still differentiate themselves by gradually changing distribution towards an endpoint, rather than all ocupying the same dense region and then taking different large jumps to their respective stressed states. This may not be necessary if we observe that the economy does tend to bounce within a normal region and then jump to a stressed region. Should I include a figure showing that the mean values of the points in diffusion space all start to go towards the central high density region? Maybe a density cap would be a good idea so that we have generally dense regions, where the path isn't directly forced right into the most dense part.

We now begin to compare the Markov bridges to the density-aware path in diffusion coordinates. In Figure 17 we can see that increasing α decreases the expected distance from the Markov bridge I am not sure if I am using the correct terminology; is a Markov bridge a single path or is it the entire distribution? to the density-aware path. This is an encouraging result as it suggests that our density-aware path may give us a meaningful path Why? What would it mean if this was the case? Maybe discuss the limits as $\beta, \alpha \to \infty$. Maybe discuss how the means of the paths may converge towards eachother in some conditions. It looks like.

NOW USING MEAN DENSITY SO FOLLOWING PARAGRAPH IS USELESS In Figure 18 we present the main findings of this section Is this the main finding? What about tying it to the density-aware paths? that increasing α increases the average minimum density of Markov bridges in diffusion space. This is a direct analogy to the discrete density-aware path with sparsity avoidance. Now the transitions much prefer jumping to nodes with high density. We can

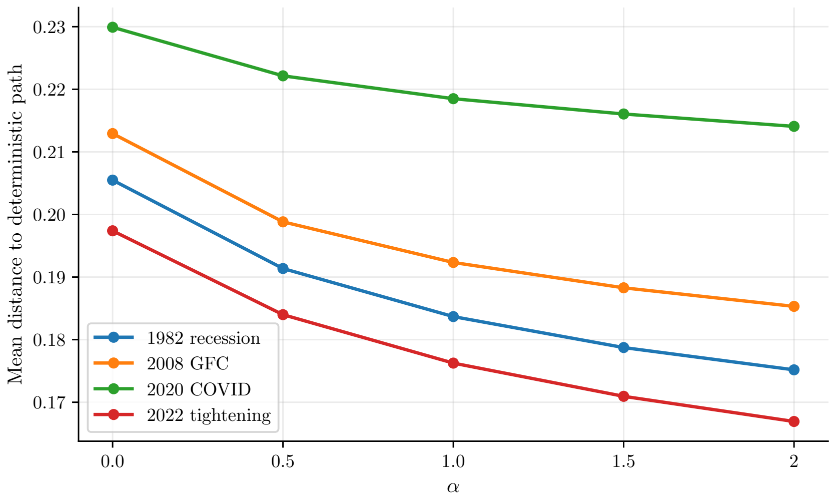


Figure 17: Caption

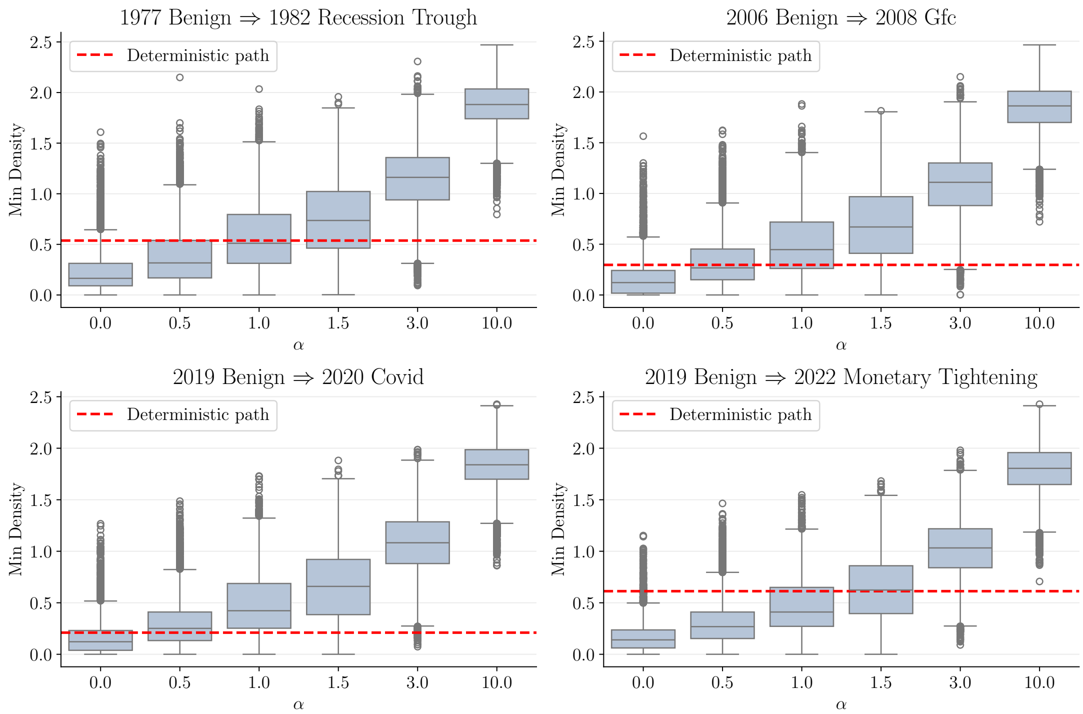


Figure 18: Caption

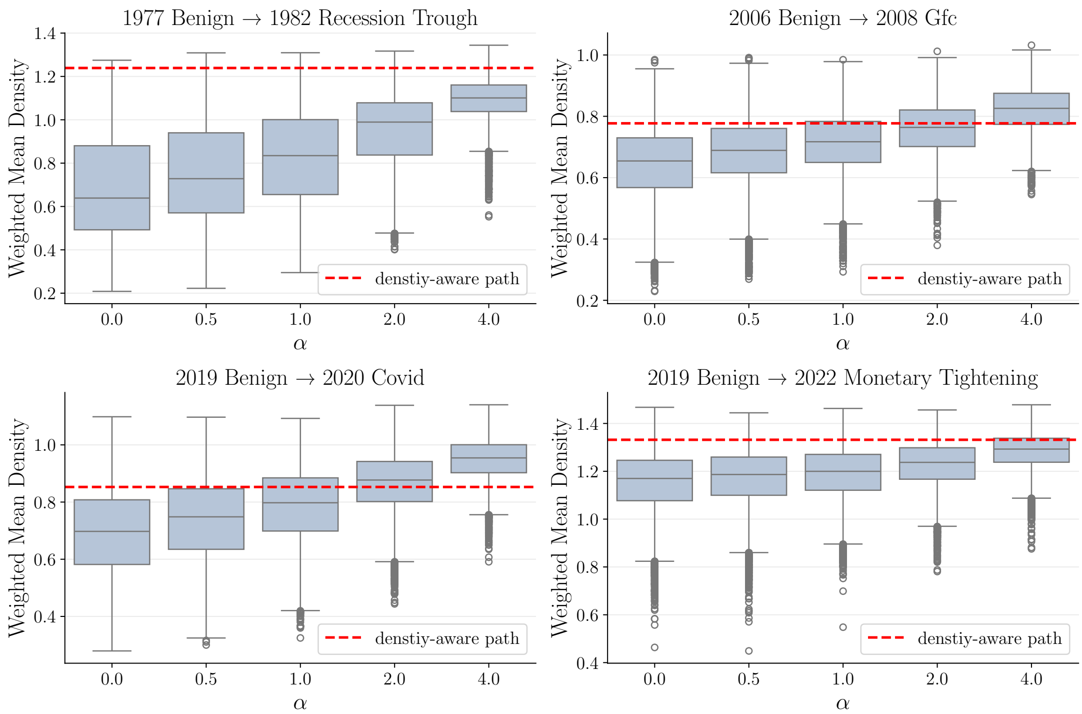


Figure 19: Caption

see an interesting patter occur as we increase α, the range of the Q1 and Q3 quantiles increases from $\alpha = 0$ until $\alpha = 1.5$ and then decreases again for $\alpha \geq 3$. At low α the distribution is high, so lots of outliers. At high α, the distribution is tightly around a bunch of high-density nodes which may not lead to a natural path. So I would guess that for low α because the paths have such a wide distribution, then most of the paths take a very natural path through the data field which involves low density nodes, but because there are lots of high-density nodes (by definition/construction), then if a path lands up in a high density region, then it is likely to stay there, because the high-density points are highly connected and thus have a high transition probability. So most paths can find their way through a small number of paths through low-density nodes (because there are not many of these paths the box is tight). The paths that don't go this way jump to a high-density region and end up staying there thus bringing the minimum density up. On the other hand, for high values of α, the paths are forced into the highest density region possible which may correspond to an unnatural path. There are no outliers above the highest level of α box plot because those nodes are in the mean, you can't get higher density than the nodes in the paths now. The bottom outliers may represent a more natural path through a moderately dense region. This would then mean that the points are given high probability based on two things: the density and the ability to get from that node to the stress node. It may be possible that the highest density nodes are far away from the stress points and therefore, despite being highly connected, do not actually make good intermediate nodes to the stress point. Then the natural paths in lower density regions start to become outliers, giving the lower density below the Q1 tail. This does not explain why the box is tighter for high density. The variance of the minimum density for $\alpha = 1.5$ may reflect the density penalty and the natural paths coinciding, so the paths are initially brough into the higher-density region by the density penalty, but once in there, the density penatly is low enough to let the paths be made up from a wide variety of routes. The 'natural path' and 'good connecting nodes' arguments seems to make sense, because the outliers seem to be concentrated at the same density level, implying that these nodes provide a natural bridging node from the low-density stress region to the high-density inner region of the path. In my mind it is obvious that because we are taking the interior minimum, the second last node to the stress path will likely be the minimum density node. Writing this down now, it actually changes some of my analysis before. - high probability to a very few low-density nodes means tight box for low α, whereas lower probability to many medium density connecting nodes gives lots of outliers... For high α, lots of the paths are forced to jump straight to high-density nodes but there remain a lot of good connecting nodes in the medium density region which are close/connected enough to have high transition density, and high enough density to no be negatively impacted by the density penalty.

The results for the weighted mean density are similar in the fact that there is a monotonic increase in the weighted mean density of Markov bridges with an increasing α. However, the boxes also get strictly tighter as α increases. This will simply be reflecting the fact that increasing α means that the paths have less entropy (as seen in Figure 16), so the mean value of the density becomes less varied. Should I show a figure for this? Maybe just for one event? they look very similar to Figure 18

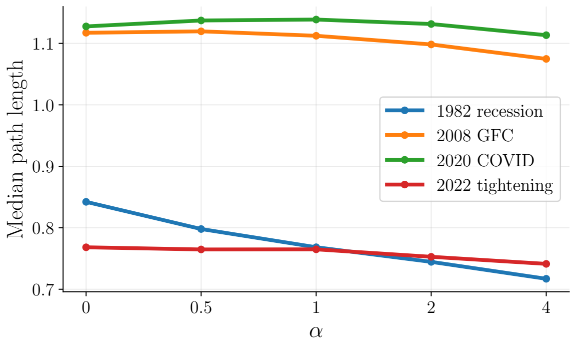


Figure 20: First caption

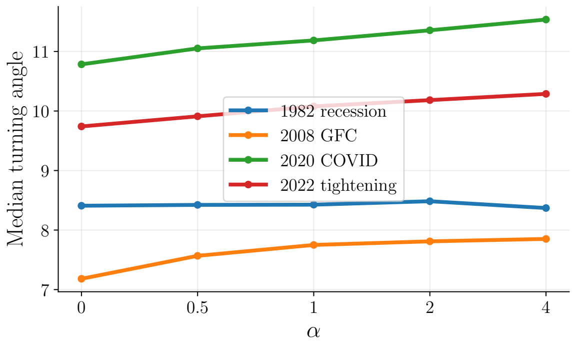


Figure 21: Second caption

I have not discussed the actual level of the average min and mean densities. It crosses over the min and mean density of the density-aware paths. This may be intersting to talk about As for the median path length (seen in Figure 20), we see a monotonic decrease with an increasing α. Quite unsure about this. It would make sense to me that forcing high density would mean that paths are longer This may be explained by the paths now taking a more direct route through the graph. However, looking at the median total turning angle, this may not be a good explanation. The median total turning angle generally increases with α (Figure 21), suggesting that paths do not become smoother or more direct. Upon visual inpsection of the 3D plots, plotting different combinations of the 4 variables, it became obvious that even though the density-aware bridges may head in the wrong direction to go through a high-density region, the fact that it was a very direct path to this high-density region and once in the region, the paths steps were not large, until they left the region with a big straight jump again. On the other hand, the unpenalised mean of the bridges took a much less direct route with larger average gaps between nodes check this. Seems as if the max gap is actually shorter for density-aware bridge means. This doesn't make sense to me thus causing the total path lengths to be larger.

Finally, we plot the distributions of the results in macroeconomic space in Figure 23 and Figure 24. We can immediately see that in Figure 23a, for FEDFUNDS, the distribution does not nearly capture the full variance of the real path. It does howevernotation capture the full variance of the linear path and the density-aware path. I suspect this is because FEDFUNDS rarely moves this much and this is a bit of a freak event (exactly what we are tying to model...?) and the 'crisis' value of FEDFUNDS is not too far out of the ordinary, so this node may be close to points in the geometry which are quite calm in comparison to the events leading up to this actual observation. This raises the fact that this model does not have a notion of time apart from in the distance calculation?? Or is this just mahalanobis distance? and so it might be geometrically reasonable for the model to take a calm path to this FEDFUNDS endpoint; however, in history, it actually took a very rough path to get here. It would be interesting to see what would happen to the variance of the distribution if we chose a very stressed point, specifically for FEDFUNDS. On the other hand, for UNRATE, we see a nice result. The confidence interval captures the real observed path, and the density-aware path (which closely follows the mean position of

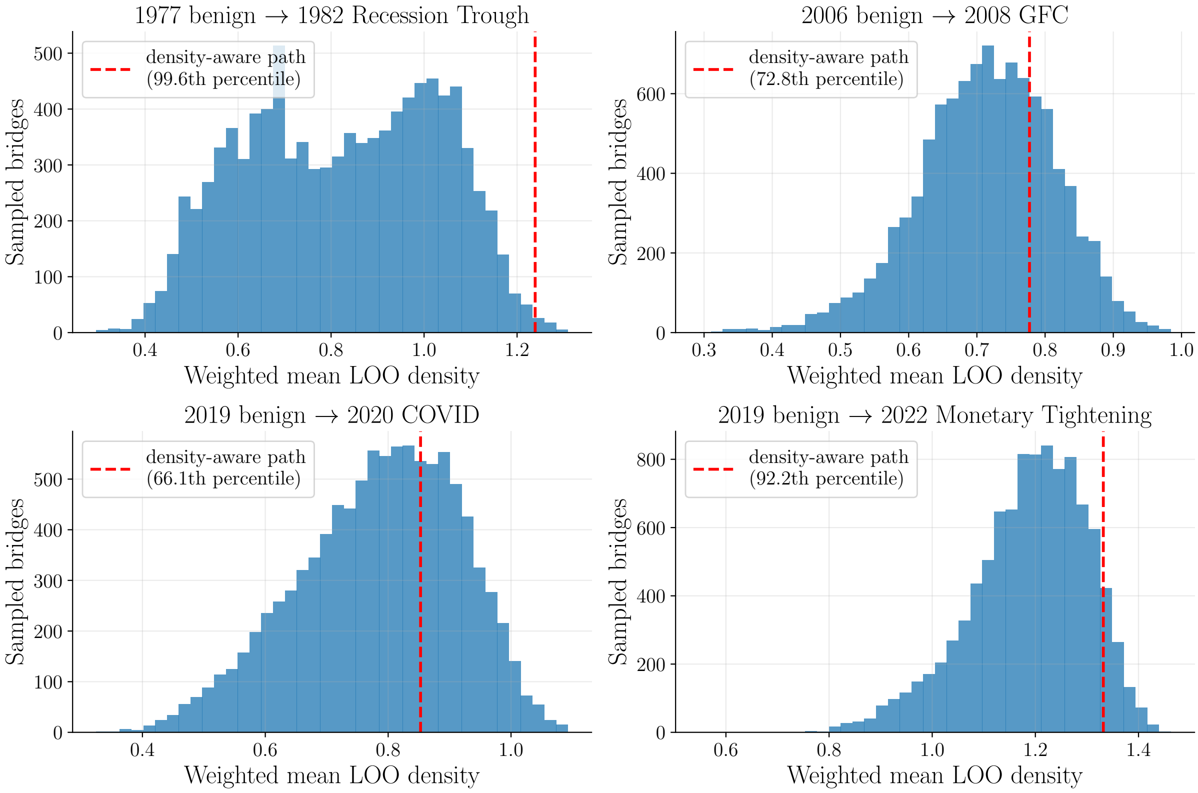


Figure 22: This is the main result for this part. Brush over the rest, I think.

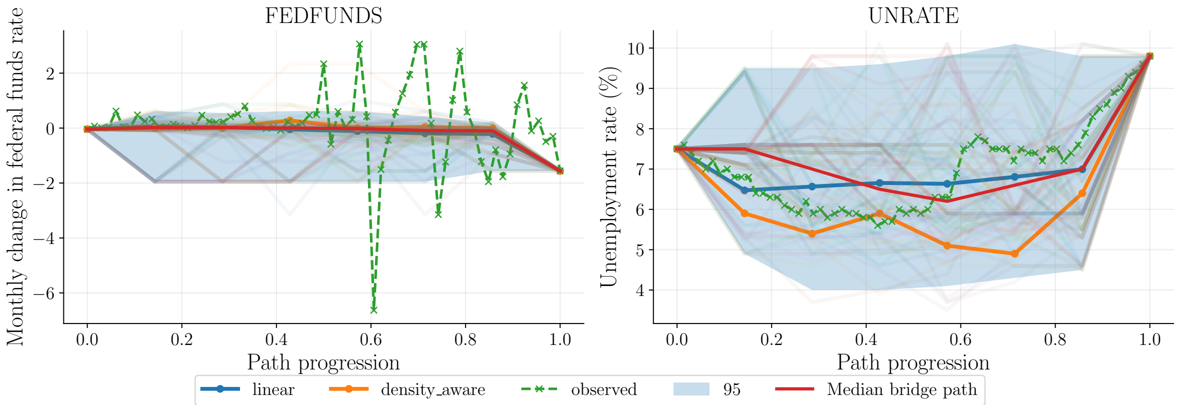
the Markov bridges). the shape of the confidence interval is also varied,, suggesting that the distribution explores different regions of the geometry as it moves forward in the path, this is something which could be important because of the different interpretations of the different regions. Perhaps the path is going through a meaningful transition through the geometry.

We see that in four out of eight plotted examples, the density-aware path does not stay within the 95% confidence interval of the Markov bridges. This suggests that there is a fundamental difference in what the different methods are doing. If we were to expect some sort of convergence, I would expect the density aware paths to be in this inverval 95% of the time.

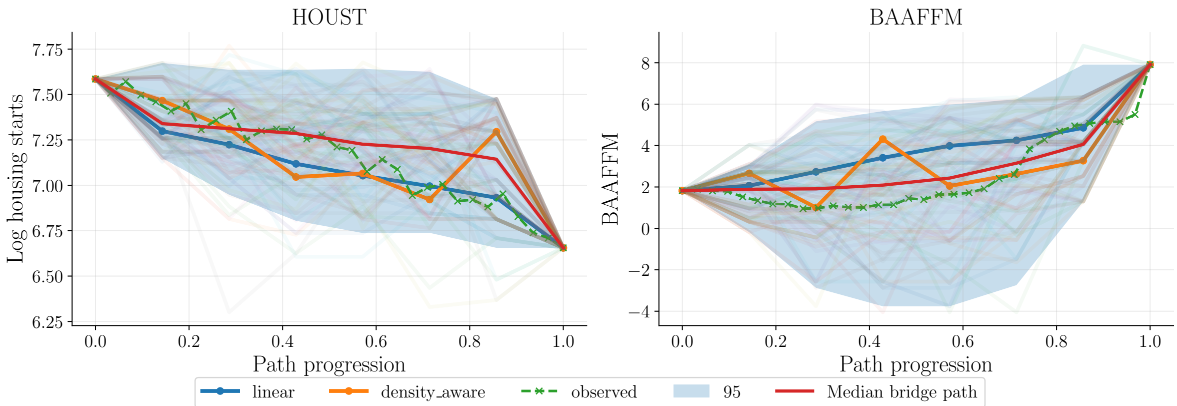
We also see that in five out of eight figures, the real path exits the 95% confidence interval given by the Markov bridges. This suggests that the distribution does not infact represent a probability observed in real life, but one specific only to this model. This is not desirable if we were to use this for modelling real assets.

Another potential issue is that for a lot of the both of the variables in Figure 23b, even though the confidence interval contains the real path, the shape of the confidence interval does not reflect the trend of the real path. This may reflect the Markov bridges jumping around a region of similar nodes in the internal part of the trajectory before quickly jumping to the stressed state. The misalignment in the trend of the confidence interval may mean that the paths are not moving through different regimes/regions of the macroeconomic space. This may result in unrealistic paths. This may be because we are constraining the paths to go to a high-density region, which may only occur during a certain regime, this would make it hard for the paths to explore new regimes between endpoints. To remidy this, it might be possible to change the density estimate for the diffusion coordinates. Instead of a gradual density increase, we could impose a maximum density which would mean that we avoid sparse regions, but we don't discriminate between regions of medium to high density. This may allow for a genuine regime shift in the Markov bridges, allowing them to more realistically move between endpoints.

Need to balance sparse region avoidance with short jumps

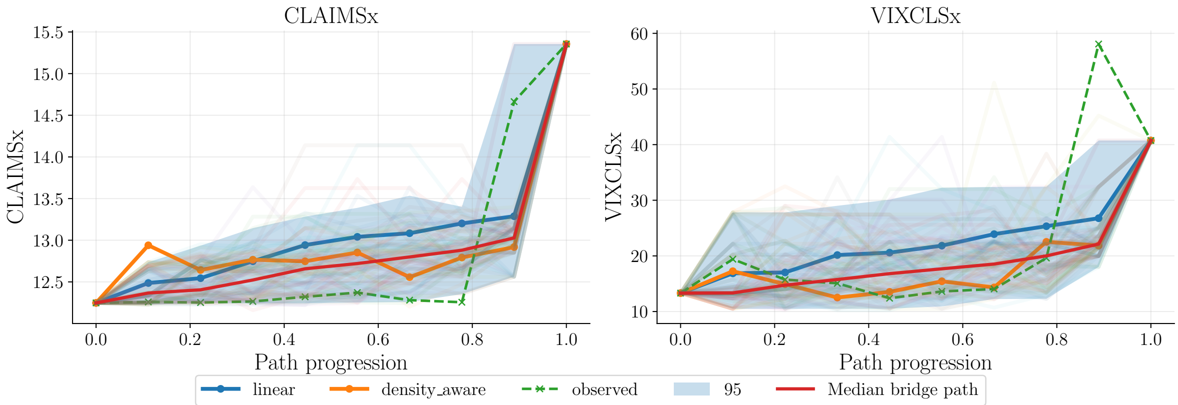


(a) Transition from the benign state in 1977 to the 1982 recession trough.

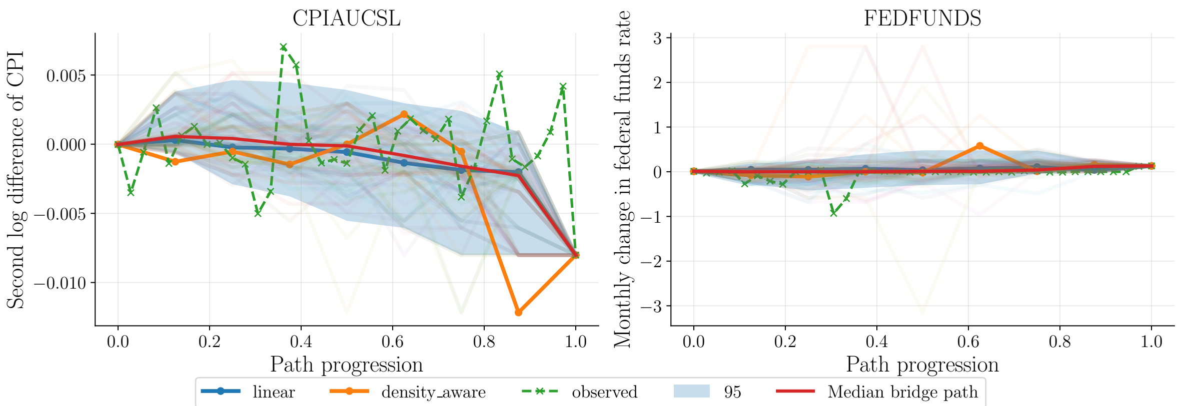


(b) Transition from the benign state in 2006 to the Global Financial Crisis in 2008.

Figure 23: Lifted macroeconomic trajectories for the early-1980s recession and the Global Financial Crisis. The graph path consists of observed transformed macroeconomic states, whereas intermediate points on the linear diffusion-coordinate path are reconstructed using local neighbourhood lifting. Plots of the 95% confidence intervals of the Markov bridges for $\alpha = 1.5$ are also included, as well as the mean position of the Markov bridge paths.



(a) Transition from the benign state in 1977 to the 1982 recession trough.



(b) Transition from the benign state in 2006 to the Global Financial Crisis in 2008.

Figure 24: Lifted macroeconomic trajectories for the COVID-19 disruption and the 2022 monetary-tightening episode. The graph path consists of observed transformed macroeconomic states, whereas intermediate points on the linear diffusion-coordinate path are reconstructed using local neighbourhood lifting. Plots of the 95% confidence intervals of the Markov bridges for $\alpha = 1.5$ are also included, as well as the mean position of the Markov bridge paths.

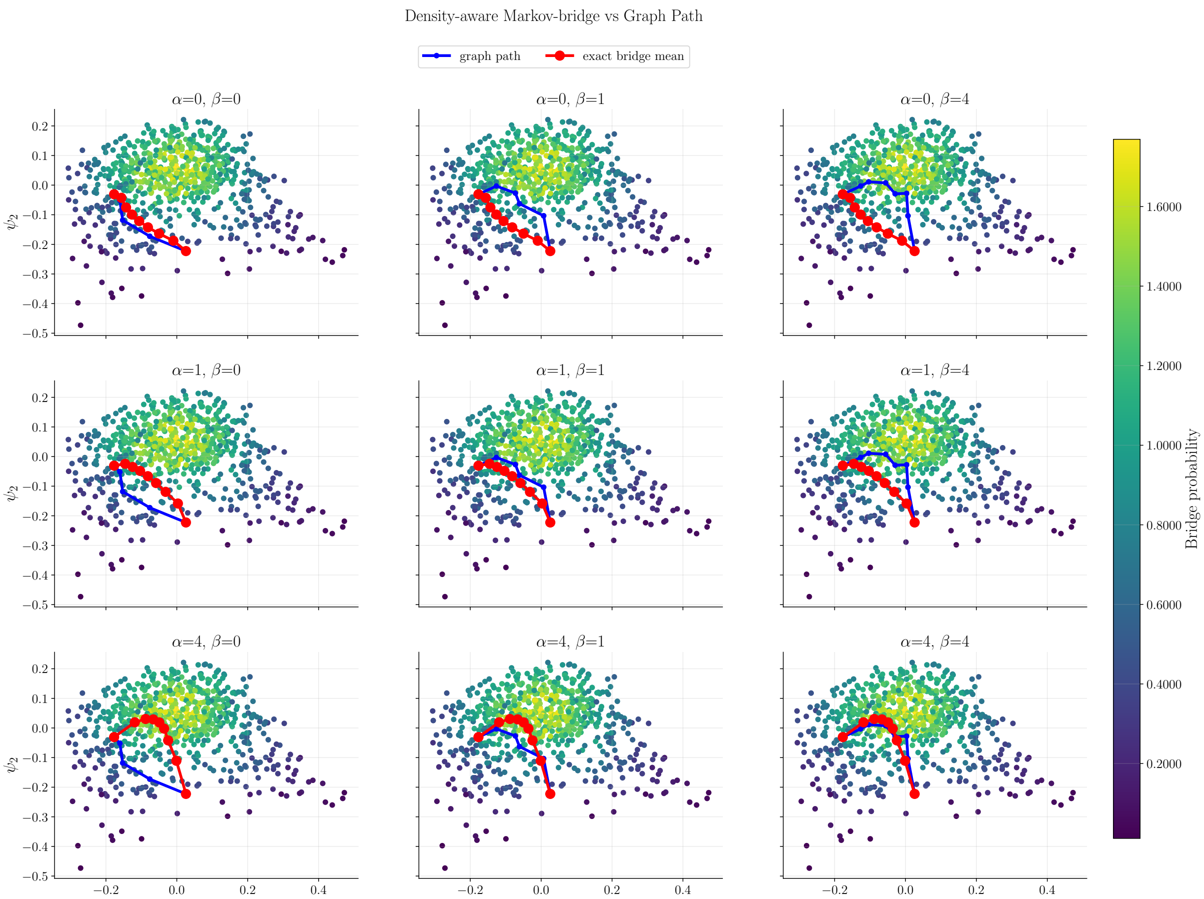


Figure 25: Caption

7 Conclusions and outlook

We could do more on the variable selection front. I basically just threw out variables which looked to be very similar. There could be an in-depth analysis done, and this would be different for every dataset. In some tests, a change in the input variables changed things a lot. Although the diffusion map seemed to be quite stable to the addition of similar variables, presumably because they get grouped into the same coordinate.

This model is quite stable to parameters; however, the interpretation of diffusion coordinates may not be. Changing the bandwidth to the median distance between quantile coordinates (median ≈ 2.5, old bandwidth $= 3$) kept the results the same, but now the choice of dimension results changed. Unlike PCA, the first eigenvectors don't necessarily give us the best reconstruction; they are ordered by the spectrum of the diffusion operator. We saw that we got not much reconstruction variance gain by adding the new fourth coordiante, but we got lots by adding the new fifth. The story in the separation was similar, where the new fifth coordinate separated near-collision pairs, whereas the new fourth coordinate didn't. The economic interpretations of the coordinates swapped around too, this suggests that actually, the eigenvectors remained similar, but the magnitute of the eigenvalues swapped, causing the new fourth coordinate to represent the same mode as the old fifth coordinate. We checked this and found a 0.9 correlation between the old fourth and new fifth, confirming our suspicions. In the initial spectrum, the magnitute of the fourth and fifth eigenvalue were very similar, so it makes sense that the ordering would be sensitive to changes in bandwidth.

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Appendices

A First appendix

A.1 Principal component benchmark methodology

To provide a conventional linear dimensionality-reduction benchmark for the proposed manifold-based stress paths, principal component analysis (PCA) was applied to the same quantile-standardised macroeconomic observations used to construct the diffusion map. The first $r = 4$ principal components were retained is this a low number?, matching the dimension of the diffusion embedding and thereby allowing the comparison to focus on the distinction between a linear PCA representation and the nonlinear diffusion embedding rather than on the number of retained coordinates.

Let

$$\Phi_{\text{PCA}} : \mathbb{R}^D \to \mathbb{R}^r$$

denote the fitted PCA transformation and let

$$y_i = \Phi_{\text{PCA}}(z_i)$$

be the PCA coordinates of observation z_i. For each of the start–stress event pairs introduced in Section 4.5, a linear path was constructed between the corresponding PCA coordinates,

$$\gamma_{\text{PCA}}(s) = (1 - s)y_{\text{start}} + s y_{\text{stress}}, \qquad s \in [0, 1].$$

The path was discretised using the same number of points as the corresponding density-aware graph path. This choice affects only the numerical representation of the continuous PCA interpolation and provides a comparable resolution for the subsequent diagnostics and plots.

Each interpolated point was mapped back to the quantile-standardised macroeconomic space using the inverse PCA transformation, truncated PCA is not invertible. It's pseudo-invertible

$$\widehat{z}_{\text{PCA}}(s) = \Phi^{-1}_{\text{PCA}}\big(\gamma_{\text{PCA}}(s)\big).$$

Since only the first four principal components are retained, this inverse transformation reconstructs observations within the rank-4 PCA subspace and does not in general reproduce the original event endpoints exactly. The PCA endpoints were therefore left as their genuine rank-4 reconstructions rather than being manually replaced by the observed macroeconomic states. The resulting loss of information was quantified for each event using the mean endpoint reconstruction error

$$E_{\text{end}} = \frac{1}{2}\left(\|\widehat{z}_{\text{PCA}}(0) - z_{\text{start}}\|_2 + \|\widehat{z}_{\text{PCA}}(1) - z_{\text{stress}}\|_2\right).$$

This is inconsistent with what you did with LNL. Don't do this, or explain why you do. Explanation at the time was that I was comparing PCA to density-aware graph paths instead of PCA vs. LNL. I have actually changed this in the code/results. This provides a direct measure of how accurately the retained PCA representation captures the particular benign and stressed states used in the scenario analysis.

The location of the PCA path relative to the empirical macroeconomic data cloud was assessed using the nearest-neighbour diagnostic introduced in Section 4.7. For an interpolated PCA reconstruction $\widehat{z}_{\text{PCA}}(s)$,

$$d_{\text{NN}}\big(\widehat{z}_{\text{PCA}}(s)\big) = \min_j \|\widehat{z}_{\text{PCA}}(s) - z_j\|_2.$$

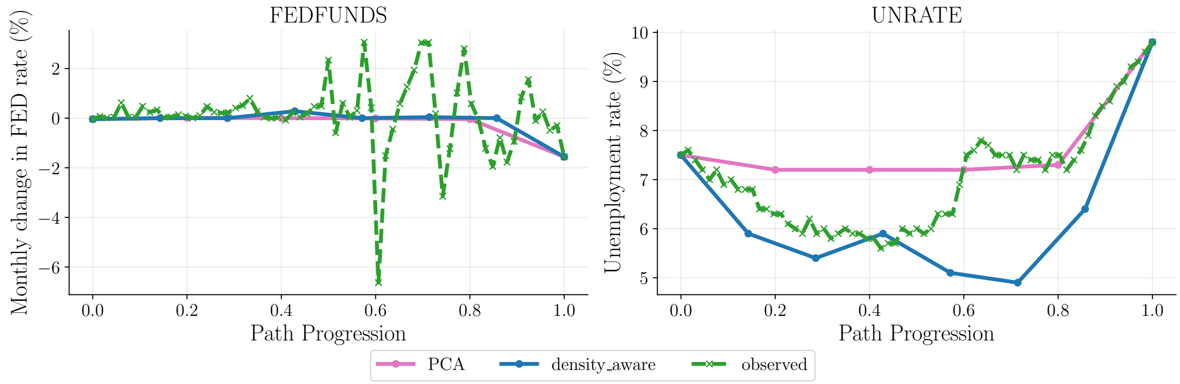
Distances were calculated in quantile-standardised macroeconomic space and compared with the leave-one-out nearest-neighbour distances of the historical observations. The median, 95th percentile and maximum path distances were recorded. As in the lifting analysis, this diagnostic is not interpreted as a measure of economic plausibility; rather, it identifies whether the generated path passes through regions unusually distant from the observed macroeconomic sample.

For economic interpretation, the PCA reconstructions were subsequently mapped to the original units of the macroeconomic variables using the inverse fitted quantile transformation. Selected variables were then plotted along the PCA path and compared with the proposed density-aware graph path and the realised historical trajectory for each event. The plotted variables were chosen to reflect the macroeconomic quantities most strongly associated with the corresponding historical shock. Since the density-aware graph path consists entirely of observed diffusion-map nodes, its macroeconomic values are taken directly from the corresponding historical observations and require no inverse mapping. This technically is an inverse mapping

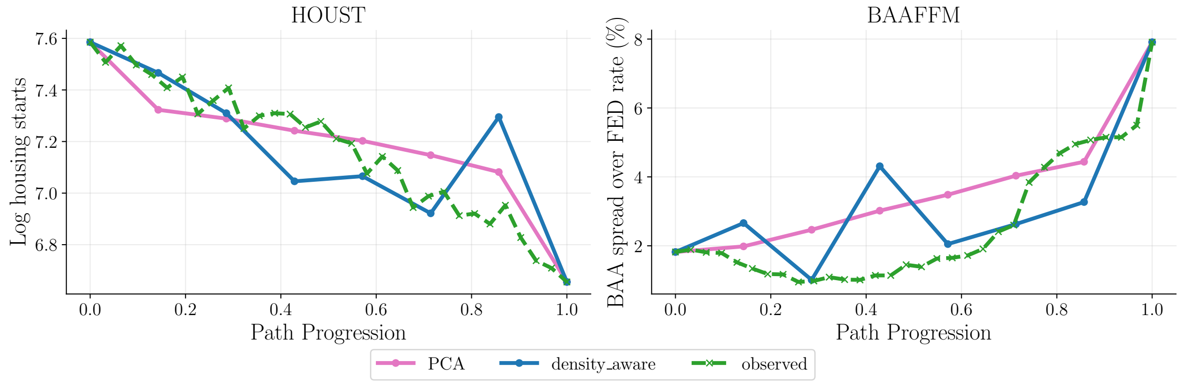
The PCA experiment therefore provides a method-level benchmark between a linear low-dimensional interpolation and the proposed nonlinear, density-aware graph construction. The endpoint reconstruction errors, nearest-neighbour diagnostics and macroeconomic trajectory comparisons are reported in Section A.2.

A.2 Principal component benchmark results

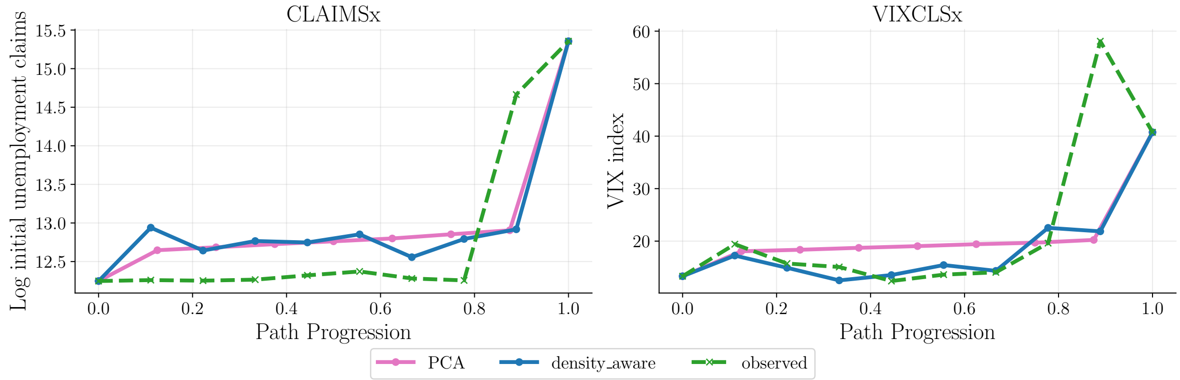
We can also compare to PCA paths. HOW DO WE CORRECTLY CHOOSE THE RIGHT DIMENSION FOR PCA? Right now I choose r=4 because that's the same as for diffusion map. We see in Figure 26a that the PCA paths are very straight. This is expected and is quite a rudamentary comparison. The rest of the figures are in Appendix C. An exploratory PCA implementation indicated that graph-constrained paths may also produce coherent routes under a linear embedding. This suggests that part of the improvement over direct interpolation arises from graph-based support constraints, rather than solely from the nonlinear diffusion representation. A systematic comparison of alternative representations is left for future work.



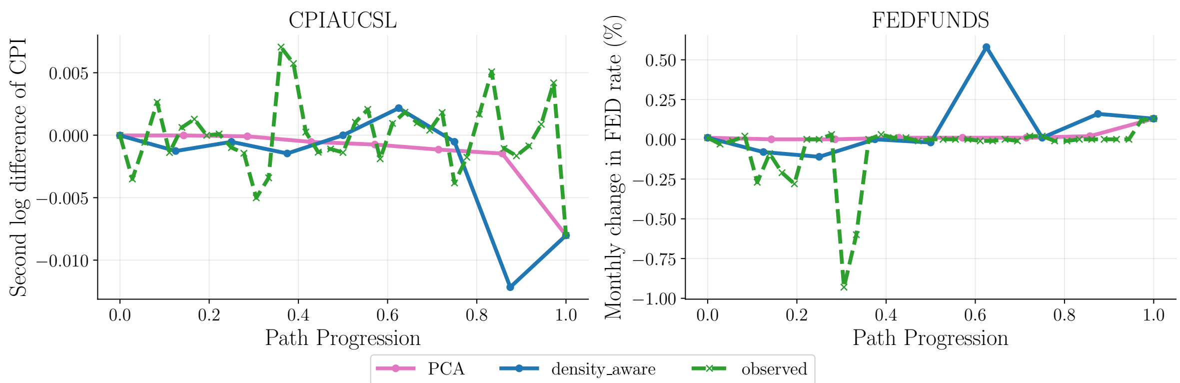
(a) Transition from the benign state in 1977 to the Global Recession in 1982.



(b) Transition from the benign state in 2006 to the Global Financial Crisis in 2008.



(c) Transition from the benign state in 2019 to the COVID-19 disruption in 2020.



(d) Transition from the benign state in 2019 to the monetary-tightening episode in 2022.

Figure 26: Lifted macroeconomic trajectories for all stress events. The graph path consists of observed transformed macroeconomic states, whereas intermediate points on the PCA path are reconstructed using the pseudo-inverse of the PCA map.

B Second appendix

Table 7: Macroeconomic variables and transformations used in the analysis.

Variable	Description	Code	Transformation used
RPI	Real personal income	5	$\Delta \log x_t$
W875RX1	Real personal income excluding current transfer receipts	5	$\log x_t - \log x_{t-12}$
DPCERA3M086SBEA	Real personal consumption expenditures	5	$\log x_t - \log x_{t-12}$
RETAILx	Retail and food services sales	5	$\Delta \log x_t$
INDPRO	Industrial production	5	$\log x_t - \log x_{t-12}$
CUMFNS	Manufacturing capacity utilisation	2	$x_t - x_{t-12}$
UNRATE	Unemployment rate	1	Level, x_t
UEMPMEAN	Average duration of unemployment	2	Δx_t
CLAIMSx	Initial unemployment insurance claims	4	$\log x_t$
PAYEMS	Total nonfarm payroll employment	5	$\log x_t - \log x_{t-12}$
AWHMAN	Average weekly hours in manufacturing	1	Level, x_t
CPIAUCSL	Consumer Price Index: all items	6	$\Delta^2 \log x_t$
CPIULFSL	Consumer Price Index: all items less food	6	$\Delta^2 \log x_t$
PCEPI	Personal consumption expenditures price index	6	$\Delta^2 \log x_t$
WPSID61	Producer price index: intermediate materials	6	$\Delta^2 \log x_t$
OILPRICEx	Crude oil price	6	$\Delta^2 \log x_t$
FEDFUNDS	Effective federal funds rate	2	Δx_t
GS1	One-year Treasury yield	2	Δx_t
GS10	Ten-year Treasury yield	2	Δx_t
T10YFFM	Ten-year Treasury yield minus federal funds rate	1	Level, x_t
BAA	Moody's Baa corporate bond yield	2	Δx_t
BAAFFM	Baa corporate bond yield minus federal funds rate	1	Level, x_t
M2SL	M2 money stock	6	$\Delta^2 \log x_t$
BOGMBASE	Monetary base	6	$\Delta^2 \log x_t$
BUSLOANS	Commercial and industrial loans	6	$\Delta^2 \log x_t$
S&P PE ratio	S&P 500 price–earnings ratio	5	$\Delta \log x_t$
VIXCLSx	CBOE Volatility Index (VIX)	1	Level, x_t
HOUST	Total housing starts	4	$\log x_t$

Notes: Transformation codes follow the FRED-MD convention: 1: level; 2: first difference; 4: logarithm; 5: first log difference; and 6: second log difference. Where a lag of 12 is specified, the corresponding first difference or log difference is computed relative to the observation twelve months earlier. The transformations applied to `UNRATE` and `CLAIMSx` differ from the standard FRED-MD recommendations.

Table 8: Industry portfolio return series used in the analysis.

Variable	**Industry**
NoDur	Consumer non-durables
Durbl	Consumer durables
Manuf	Manufacturing
Enrgy	Energy
HiTec	High technology / business equipment
Telcm	Telecommunications
Shops	Wholesale, retail and related services
Hlth	Healthcare
Utils	Utilities
Other	Other industries

The industry portfolio series are monthly returns obtained from the Kenneth R. French Data Library and are therefore not subjected to the FRED-MD stationarity transformations listed in Table 7.

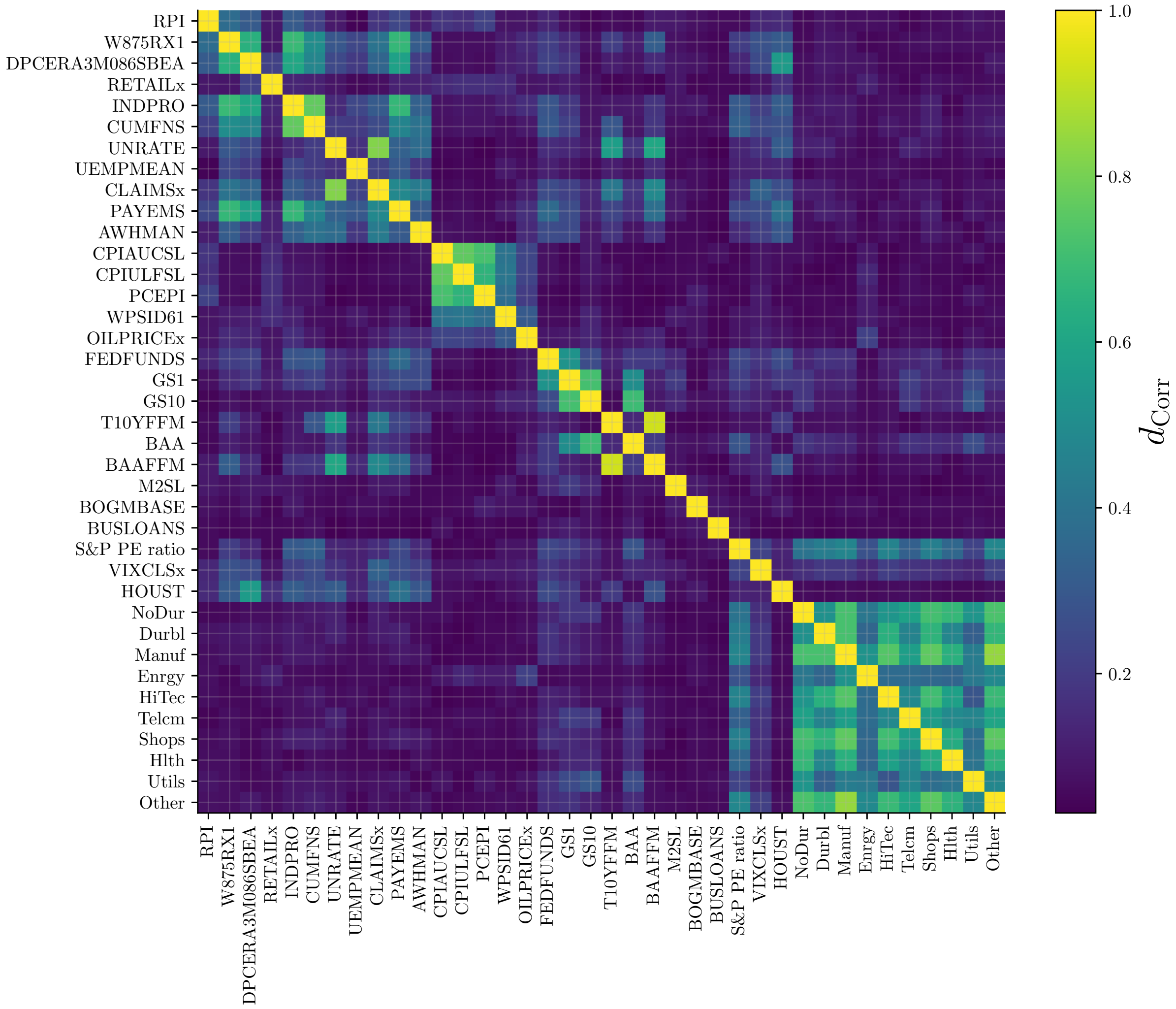


Figure 27: Distance correlation heatmap for all 38 chosen variables used to train the diffusion map

C Third appendix

C.1 Bandwidth parameter selection

Bandwidth selection results. Should I include the kernel density results?

ε	Min Degree	Degree CV
1	1.45	0.47
2	9.52	0.33
3	31.24	0.25
4	62.45	0.20
4	97.65	0.17

Table 9: Minimum degree and degree CV across bandwidths.

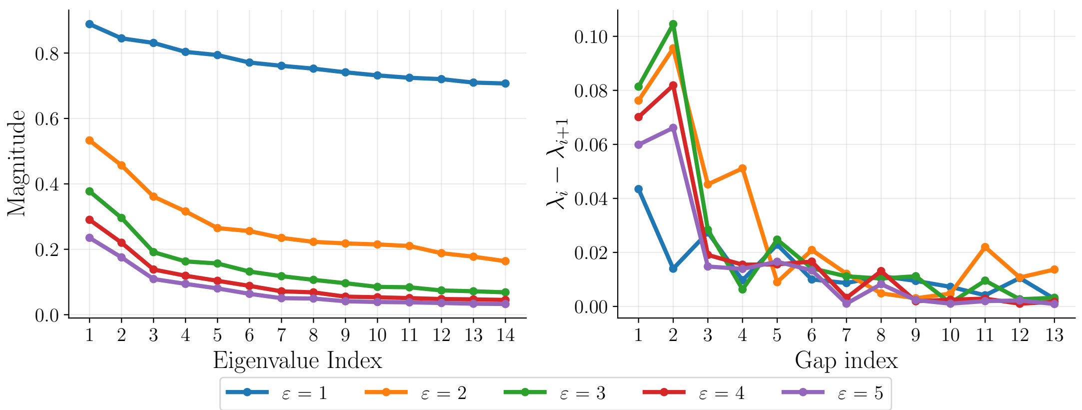


Figure 28: Caption

C.2 kNN graph construction and density estimate bandwidth

When generating the kNN graph from the diffusion coordinates using the metrics described in Section 4.4, the results over many values of k were very stable. With no clear winner, we chose $k = 15$ as our final value. The results are displayed in Table 9.

Table 10: Network Metrics by k

k	Mean degree	Graph density	Jump ratio (95th percentile)
5	6.5734	0.0089	2.2838
10	12.8668	0.0173	2.3448
15	19.3674	0.0261	2.3373
20	25.8089	0.0348	2.3475
30	38.7645	0.0522	2.3288

For the bandwidth of the graph-degree density kernel, we chose it to be one-tenth of the median squared distance between diffusion coordinates. Choosing a bandwidth that was too small resulted in many points being classified as low-density, whereas choosing one that was too large meant the density was very similar across a wide range of points. The resulting density of points was similar to nearby choices of this bandwidth. The density floor δ was chosen to be 10^{-6}; however, with the chosen bandwidth all the resulting densities were larger than this value, so it did not affect downstream results. I think that this can all just go in the methodology section. These are not really results, and there is not much to say about these choices. We just had to make them.

D Fourth appendix

In [3], they used a linear lifting operator to lift the diffusion coordinates to macroeconomic variables. The point of using a new method is to try to improve on the linear interpolation method by giving even naive linear paths more historical support. The whole point of the diss is that we are constrained to use diffusion coordinates and now we want to find a good way to model things while using them. So even the linear interpolations we want to improve.

We use the LNL method with the parameters described in Section 4.7

The results of the reconstruction tests described in Section 4.7 show that local-neighbour lifting performes slightly worse . Results are displayed in Table 11.

Table 11: Model Performance Metrics

Method	**RMSE**	$\mathbf{R^2}$
Baker linear lift	1.3176	0.4530
Local-neighbour lift	1.3209	0.4503

We also test the nearest-neighbour statistics of paths for each event for both methods. This test is to analyse the reconstructions of the paths and compare to the historical data cloud.

Table 12: Nearest-neighbour diagnostics for the lifted linear paths. Distances are measured in the quantile-standardised macroeconomic space. The ratio R_{95} compares the 95th percentile of the path distances with the 95th percentile of the leave-one-out nearest-neighbour distances among the historical observations. Values of $R_{95} > 1$ indicate that the lifted path contains points farther from the historical data cloud than is typical at the corresponding historical threshold.

Scenario	Lift	Median d_{NN}	R_{95}
1977–1982 recession	Global linear	1.04	0.66
	LNL	1.03	0.67
2006–2008 GFC	Global linear	1.15	1.06
	LNL	1.02	0.61
2019–2020 COVID-19	Global linear	1.09	0.89
	LNL	1.00	0.65
2019–2022 tightening	Global linear	1.10	0.68
	LNL	1.08	0.66

In this test we can see that the local-neighbour lift performs better than the Baker-style lift on every metric (apart from Median d_{NN} for 2019 $\rightarrow$ 2022 Monetary tightening). All of the results suggest that both of these methods work reasonable well because the median distance from path points to the observation data cloud is less than the average distance between points in the data cloud. This is not the case for the Global Financial Crisis path with the Baker-style lift, where the 95% quantile of the distance to the closest neigbour is roughly 3 compared to the historic 95% quantile of approximately 1.8. This shows that the Baker-style lifting may not be suitable for all stress scenarios, because it is sending points in diffusion space to points very far away from points in macro space. The results show that even though we are using a naïve approach of linear interpolation, we can use a modified approach to lifting unseen points to macro space, which seem to pull points closer to the observed data cloud. This may prove useful in future work. For the rest of the dissertation, when lifting linear paths to macroeconomic variables, we will use LNL.